

Related U.S. Application Data

a continuation-in-part of application No. PCT/US2019/044865, filed on Aug. 2, 2019.

- (60) Provisional application No. 63/626,355, filed on Jan. 29, 2024, provisional application No. 62/715,203, filed on Aug. 6, 2018, provisional application No. 62/841,933, filed on May 2, 2019.

(56) **References Cited**

U.S. PATENT DOCUMENTS

4,456,421	A *	6/1984	Robson		B60P 1/4428	414/541
5,062,451	A	11/1991	Vadasz F et al.			
5,164,896	A	11/1992	Nagayasu			
5,190,195	A	3/1993	Fullhart			
5,244,133	A	9/1993	Abbott			
D341,348	S	11/1993	Williams			
5,333,888	A	8/1994	Bell			
5,406,051	A	4/1995	Lai			
5,423,566	A	6/1995	Warrington			
5,526,971	A *	6/1996	Despain		B60R 9/06	224/532
5,527,146	A	6/1996	Allsop et al.			
5,579,972	A	12/1996	Despain			
5,593,172	A	1/1997	Breslin			
5,655,562	A	8/1997	Palmer			
5,685,686	A	11/1997	Burns			
5,688,804	A	11/1997	Rosen			
5,730,345	A	3/1998	Yeckley et al.			
5,845,831	A *	12/1998	Nusbaum		B60R 9/10	224/535
6,460,745	B1	10/2002	Weaver			
6,502,845	B1	1/2003	Van Vleet			
6,562,396	B1	5/2003	Katterhagen			
6,835,021	B1	12/2004	McMillan			
6,974,147	B1	12/2005	Kolda			
7,004,991	B2	2/2006	Allsop			
7,077,417	B2	7/2006	Shoemaker			
7,217,078	B2 *	5/2007	Short		B65F 1/1452	224/537
7,854,331	B2	12/2010	Ouyang			
8,066,162	B2	11/2011	Chiang			
8,950,592	B1	2/2015	Greenblat et al.			
9,027,950	B2	5/2015	Lahn			
9,056,640	B2	6/2015	Wineyand			
9,321,407	B2 *	4/2016	Loken		B60R 9/06	
9,376,063	B2	6/2016	Hein et al.			
10,293,757	B2	5/2019	Bass et al.			
10,501,023	B1	12/2019	Mayers			

10,981,487	B1	4/2021	Russo			
11,142,133	B1 *	10/2021	Oshman		B60R 9/10	
11,479,182	B1 *	10/2022	Yang		B60R 9/10	
2001/0035446	A1	11/2001	Walstrom et al.			
2003/0080159	A1	5/2003	Munoz et al.			
2003/0089751	A1	5/2003	Munoz et al.			
2005/0082329	A1	4/2005	Cohen			
2005/0129490	A1	6/2005	Hutchins			
2006/0032880	A1	2/2006	Setlemayer			
2008/0085176	A1	4/2008	Statkus			
2008/0099522	A1	5/2008	Clausen et al.			
2008/0100076	A1	5/2008	Potts			
2009/0020577	A1	1/2009	Johnson et al.			
2009/0120984	A1	5/2009	Sautter et al.			
2009/0261136	A1	10/2009	Skoff			
2010/0224746	A1	9/2010	Johnson et al.			
2010/0272548	A1	10/2010	McKamey			
2010/0290876	A1	11/2010	Conaster			
2011/0057008	A1 *	3/2011	Clausen		B60R 9/10	224/504
2011/0101060	A1	5/2011	Hammond			
2012/0048823	A1	3/2012	Kriner			
2012/0142602	A1	6/2012	Brady			
2012/0168480	A1	7/2012	Gray			
2013/0101383	A1	4/2013	Lynch, Jr. et al.			
2013/0309053	A1	11/2013	Cameron			
2013/0327802	A1 *	12/2013	Hammond		B62H 3/02	211/17
2014/0151421	A1	6/2014	Loken			
2016/0107559	A1	4/2016	Russo et al.			
2016/0129845	A1	5/2016	Mehlen			
2016/0239847	A1	8/2016	Arvapally et al.			
2016/0346141	A1	12/2016	Guertler et al.			
2018/0354427	A1 *	12/2018	Yazdian		B60R 9/10	
2019/0016271	A1	1/2019	Hintz			
2019/0161022	A1 *	5/2019	McFadden		B60R 9/06	
2019/0210675	A1 *	7/2019	Clark		B62H 5/003	
2019/0366939	A1	12/2019	Salisbury			
2020/0189481	A1	6/2020	Doelitsch			
2020/0198546	A1	6/2020	Casagrande et al.			
2020/0346505	A1	11/2020	Loewen			
2020/0406830	A1	12/2020	Owen et al.			
2021/0107580	A1	4/2021	Flynn			

OTHER PUBLICATIONS

PCT/International Preliminary Report on Patentability dated Feb. 9, 2021, for PCT International Patent Application No. PCT/2019/044965 filed Aug. 2, 2019, 8 pages.
 "ALTA Racks" Aug. 6, 2017 (retrieved on Sep. 20, 2019), retrieved from the internet at <https://www.instagram.com/p/BXdA771hUYi/>.

* cited by examiner

FIG. 1

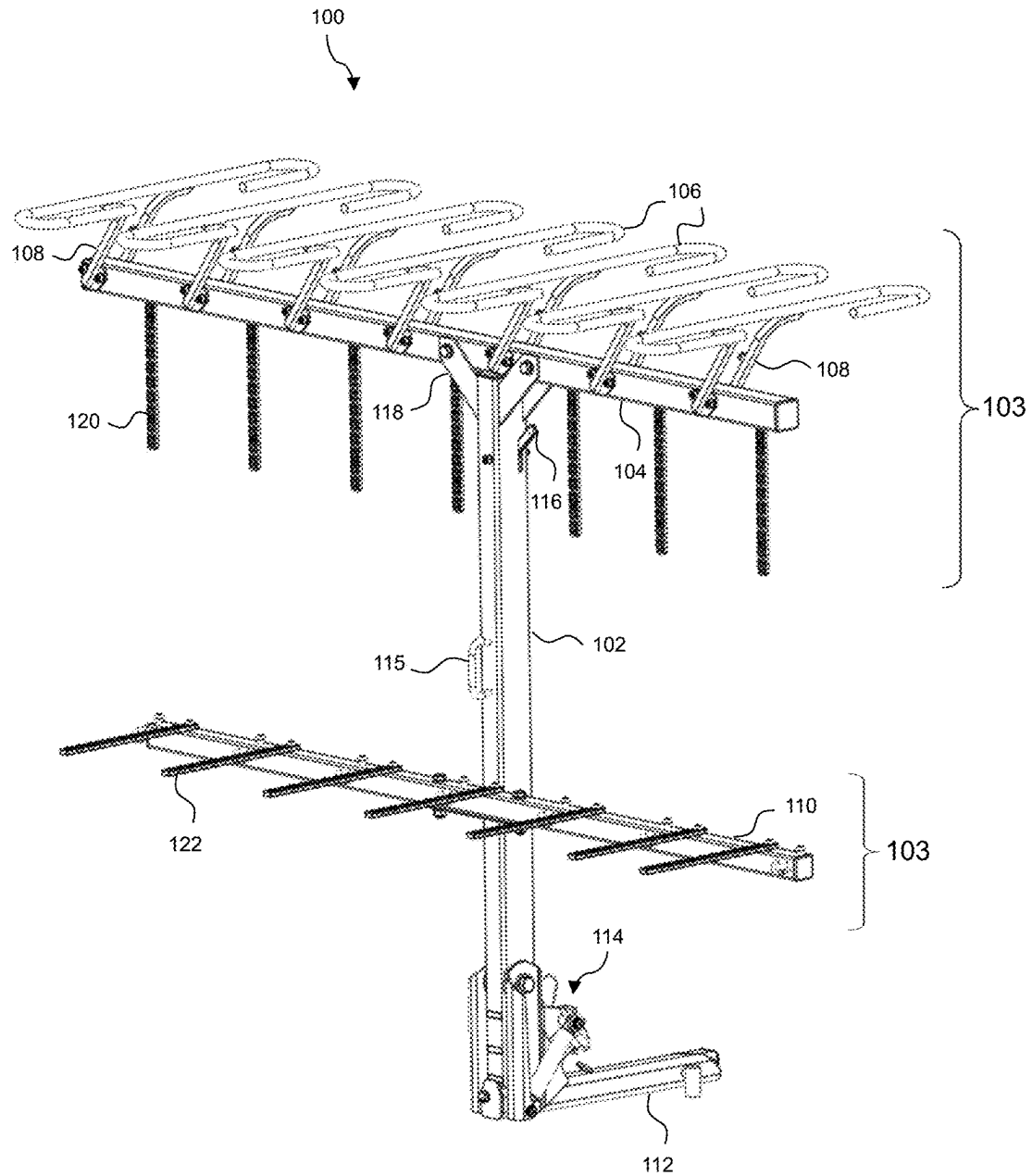


FIG. 2

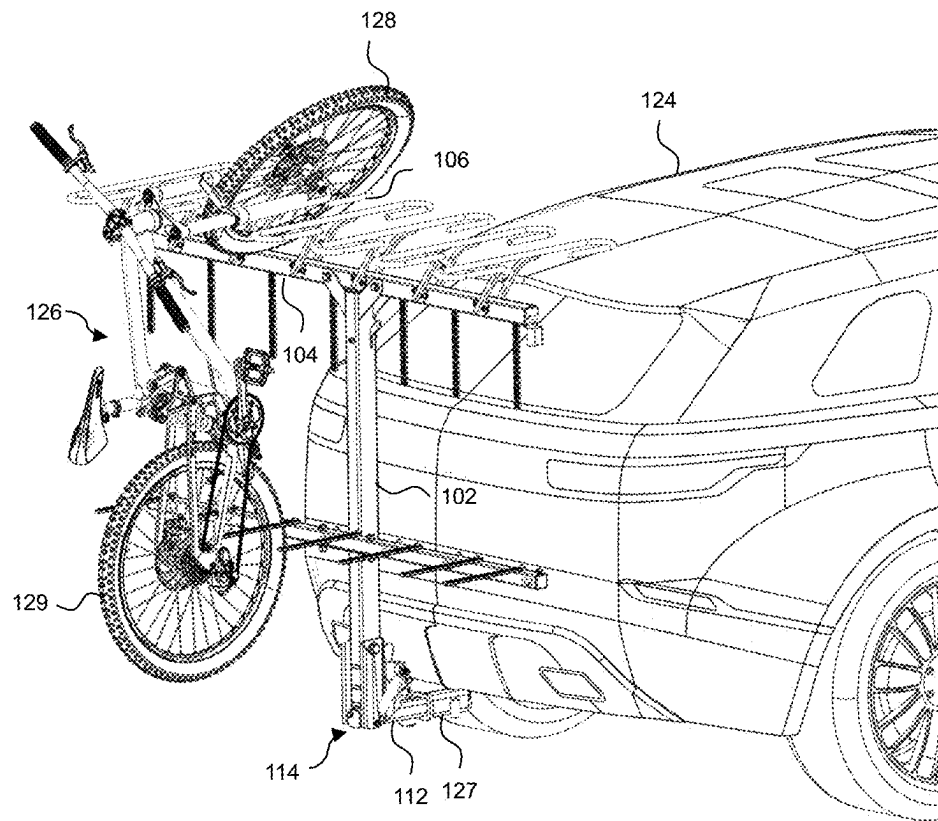


FIG. 3

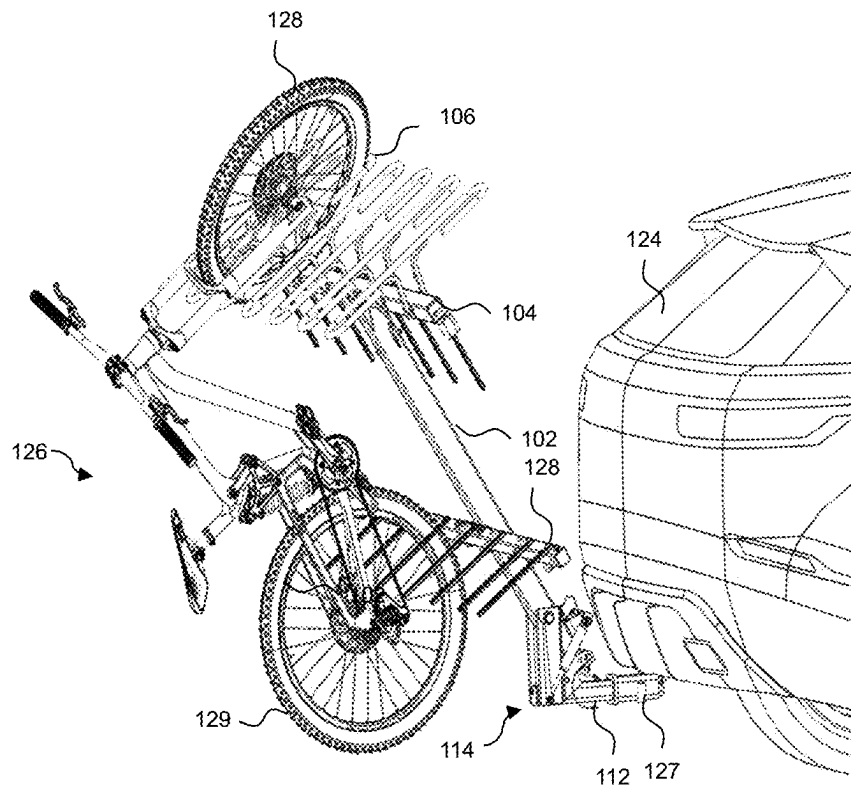


FIG. 4

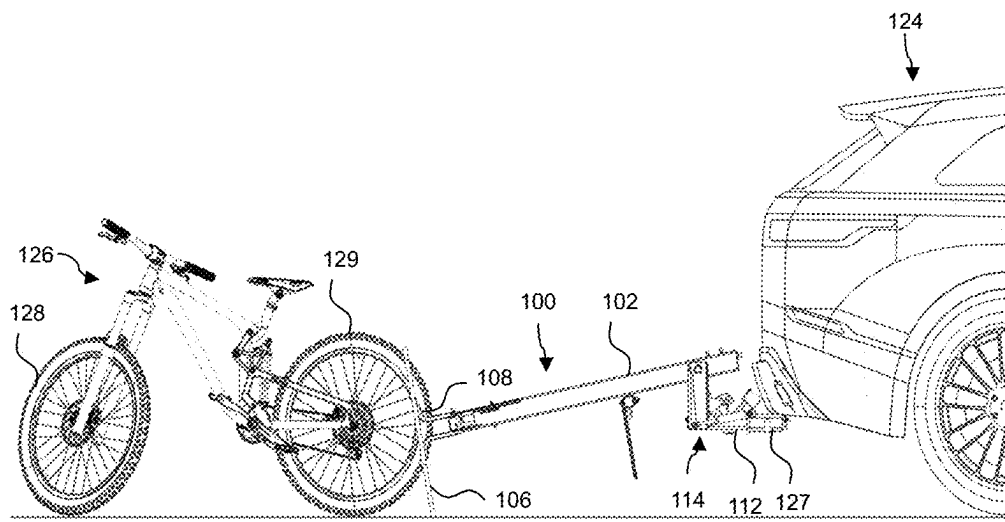


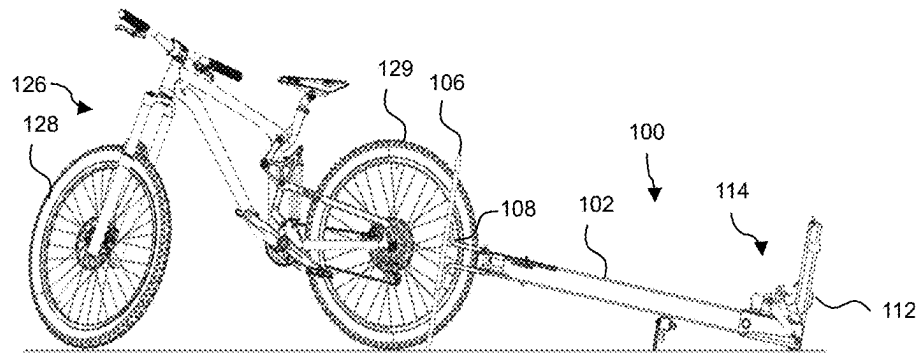
FIG. 5

FIG. 6

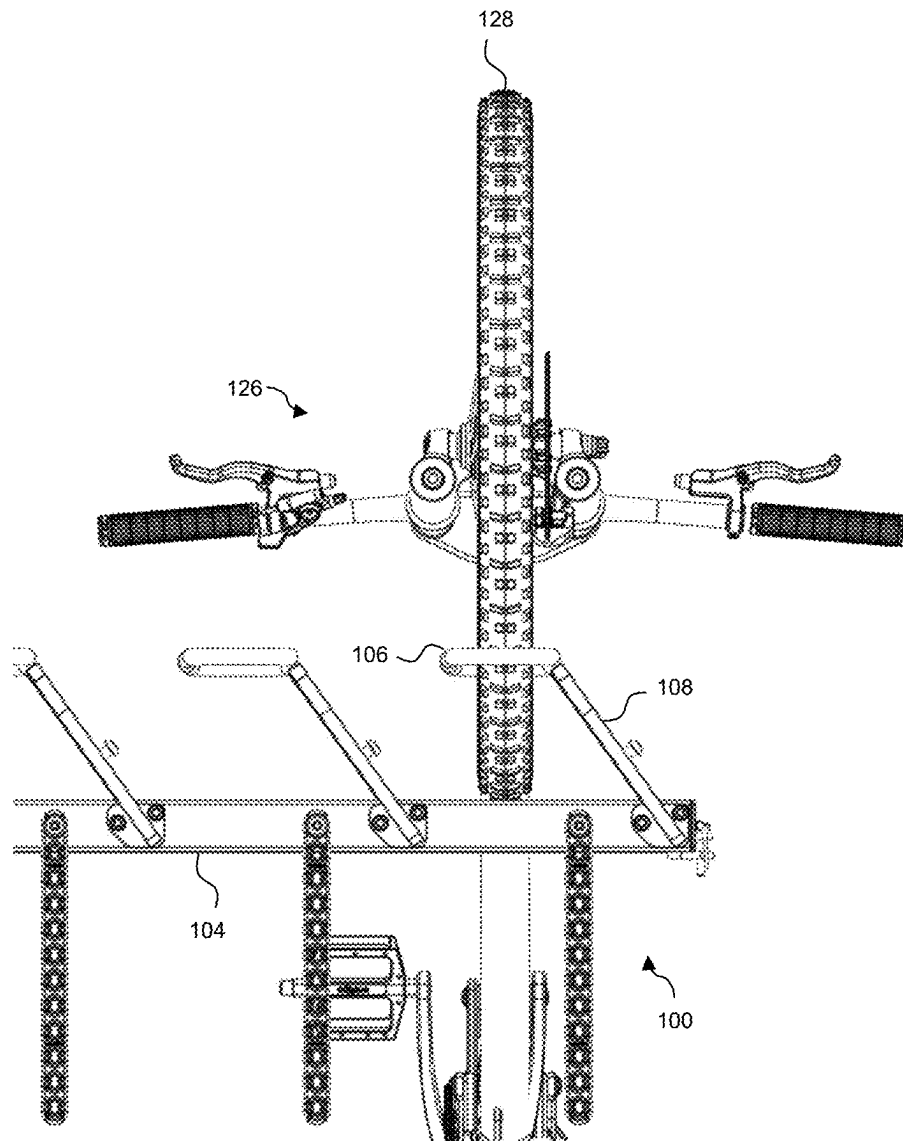


FIG. 7

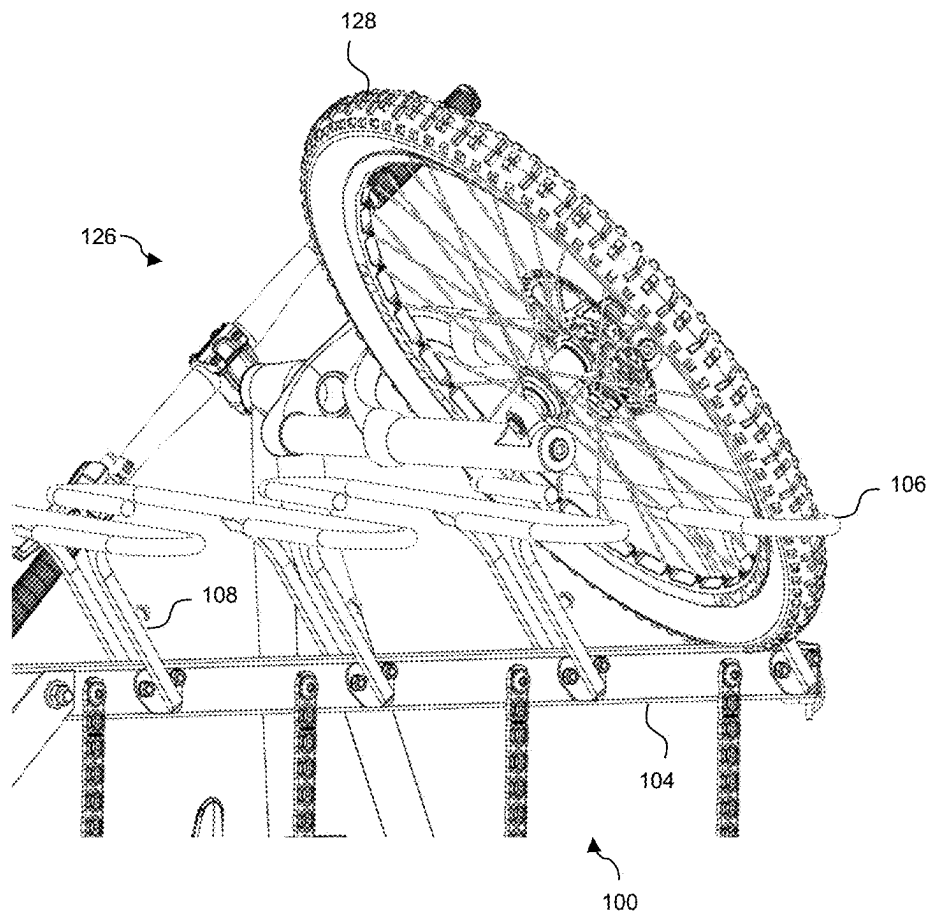


FIG. 8

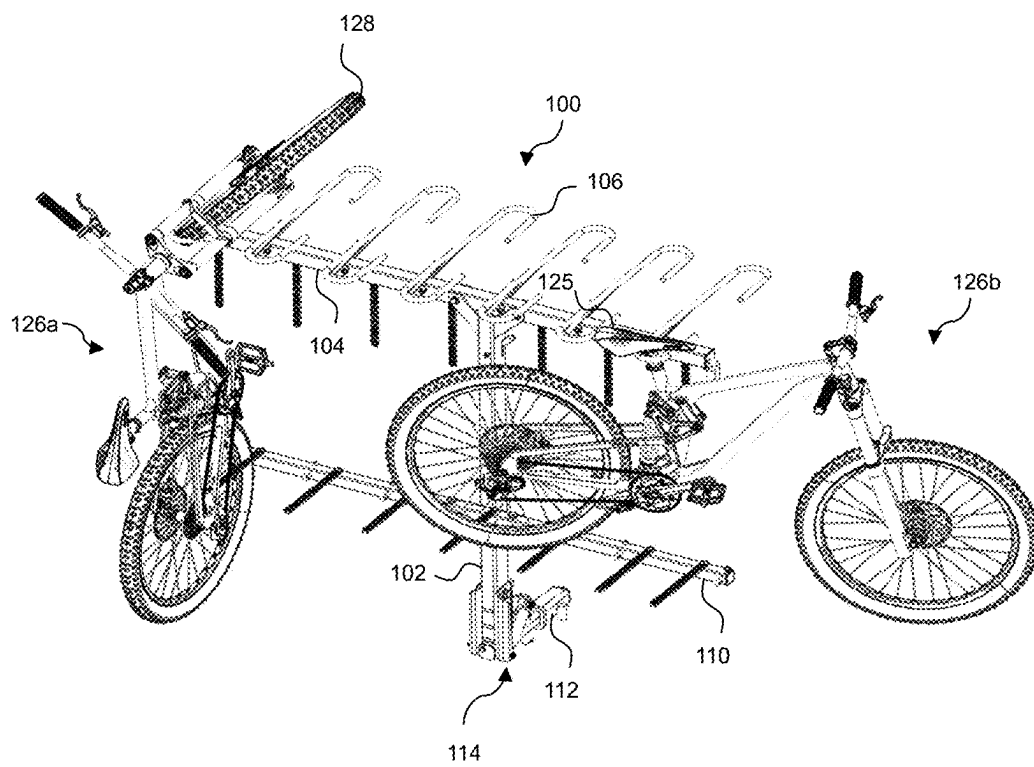


FIG. 9A

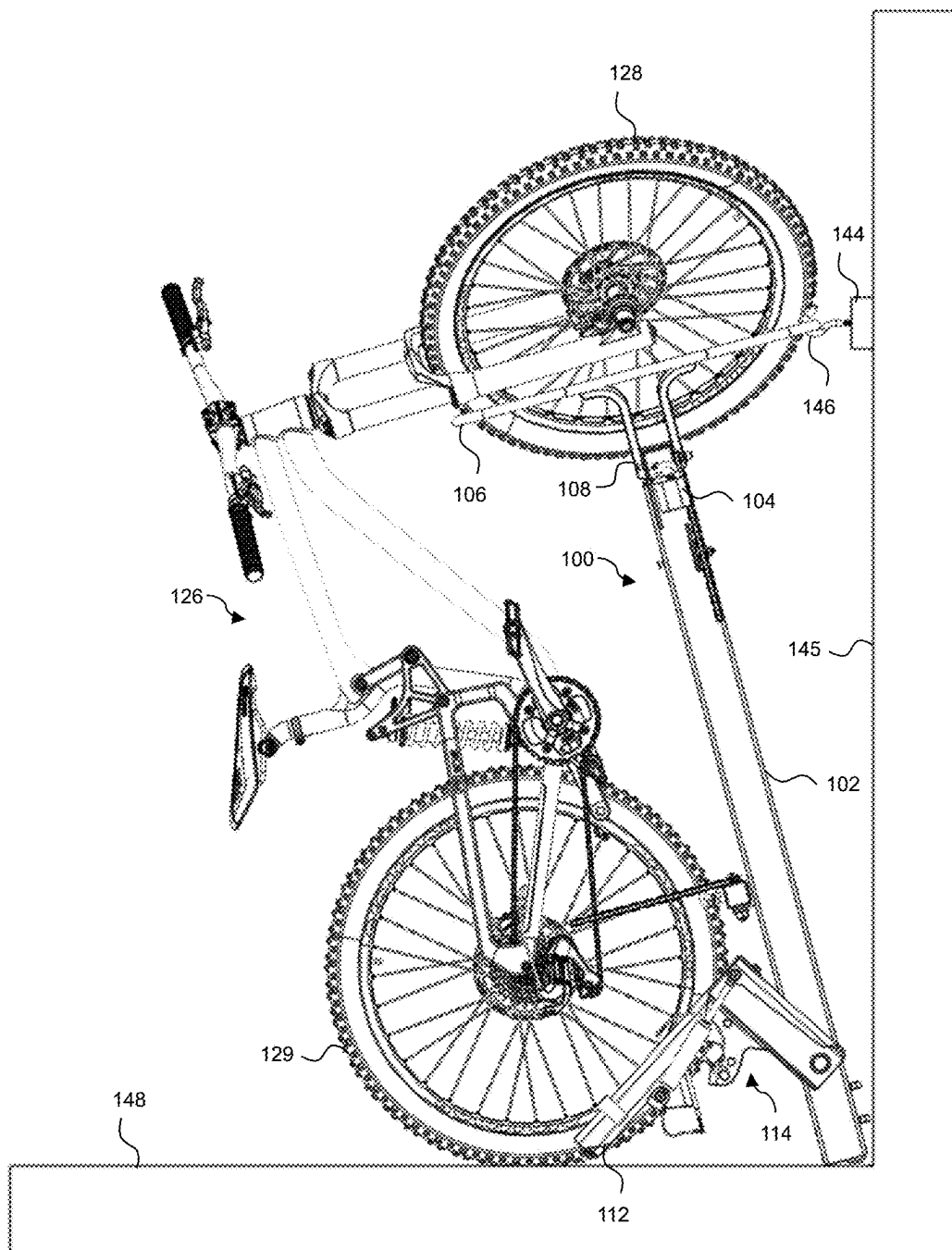


FIG. 9B

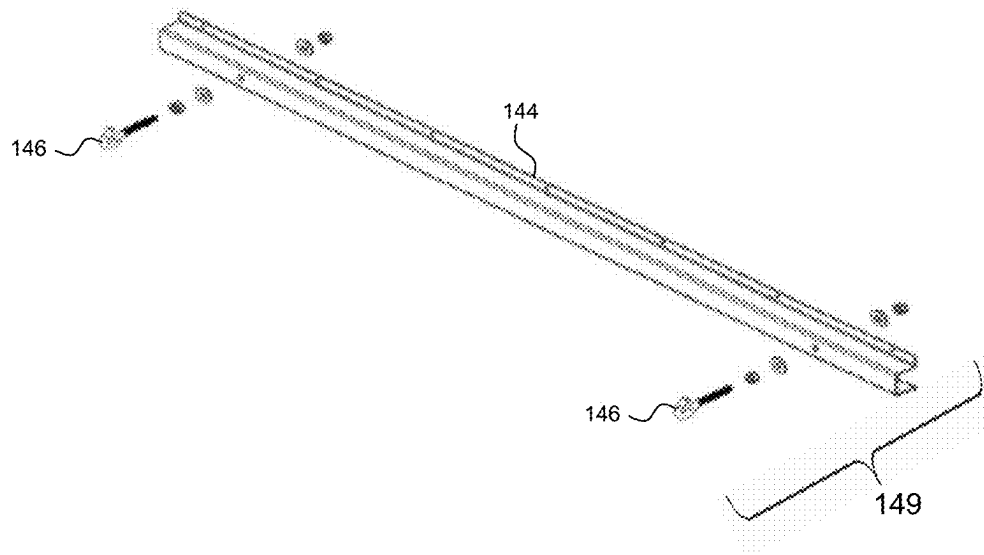


FIG. 9C

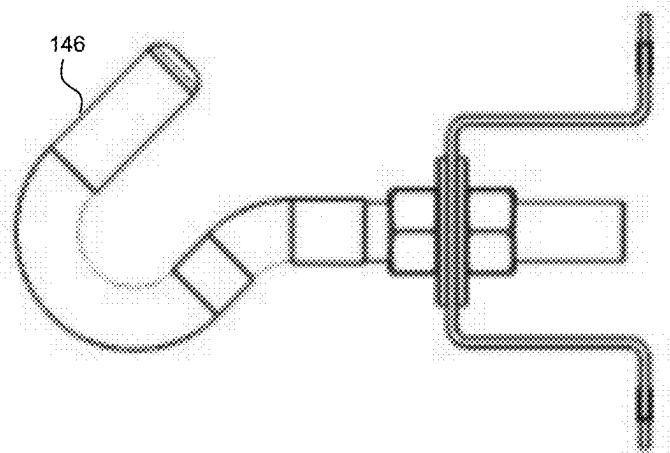


FIG. 10

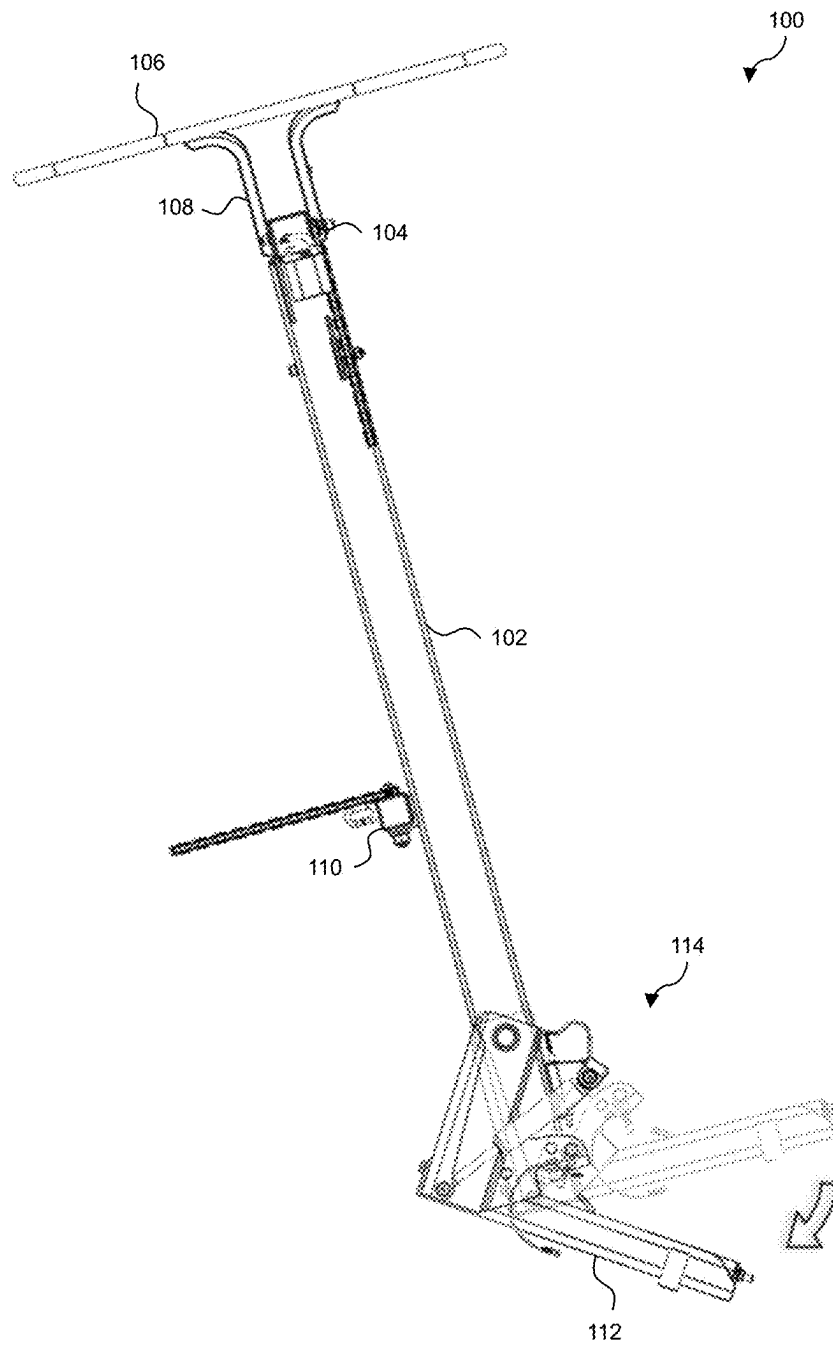


FIG. 11

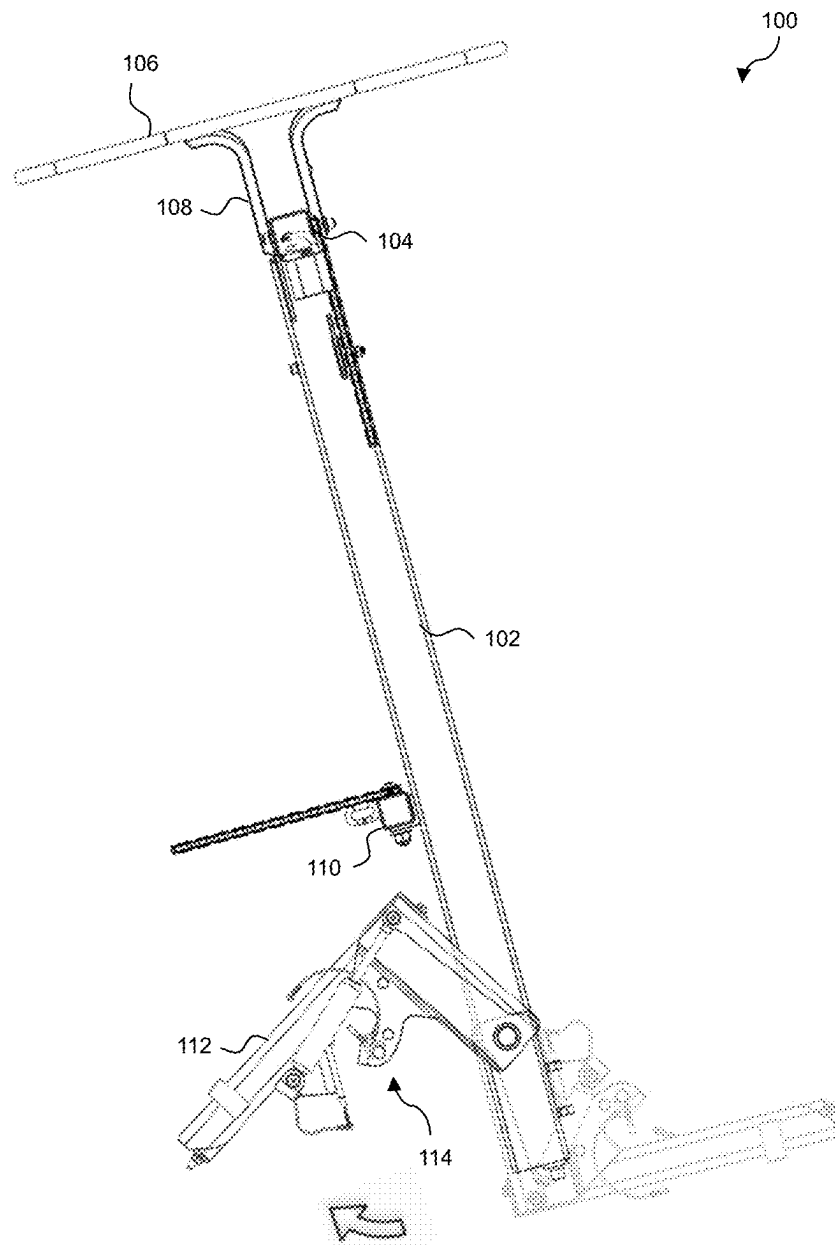


FIG. 12

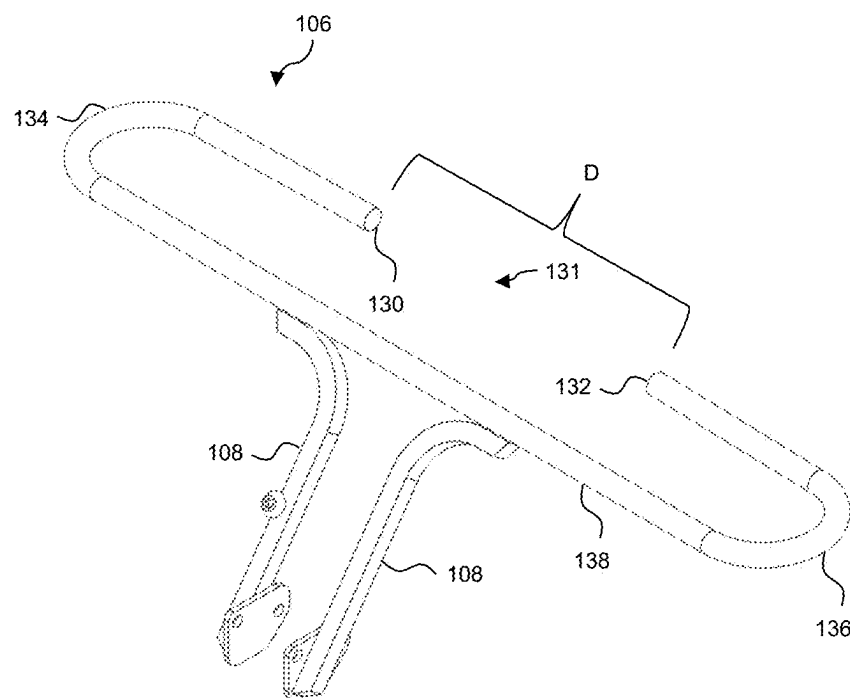


FIG. 13

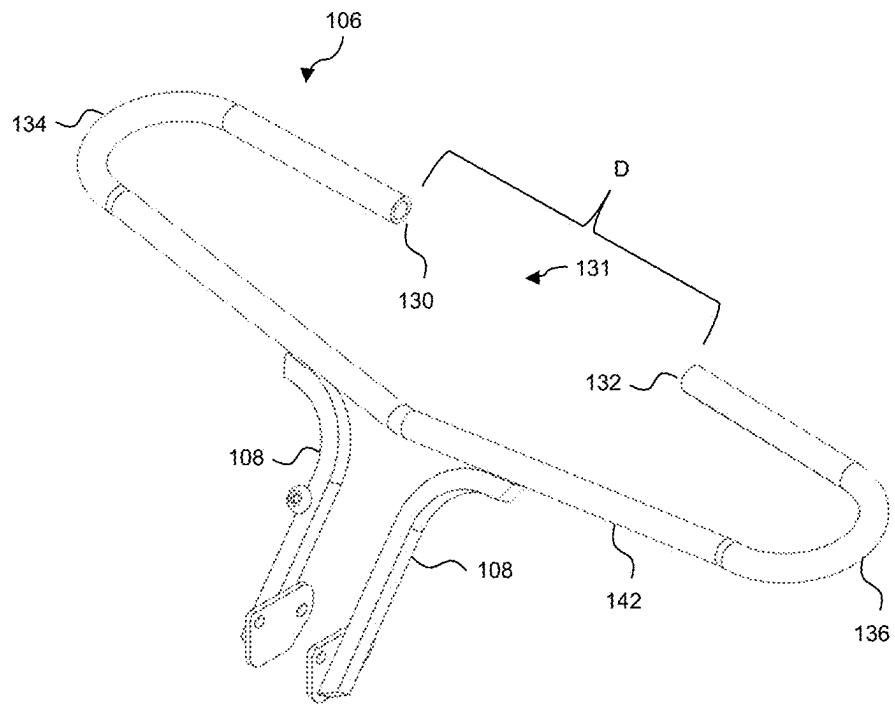


FIG. 14

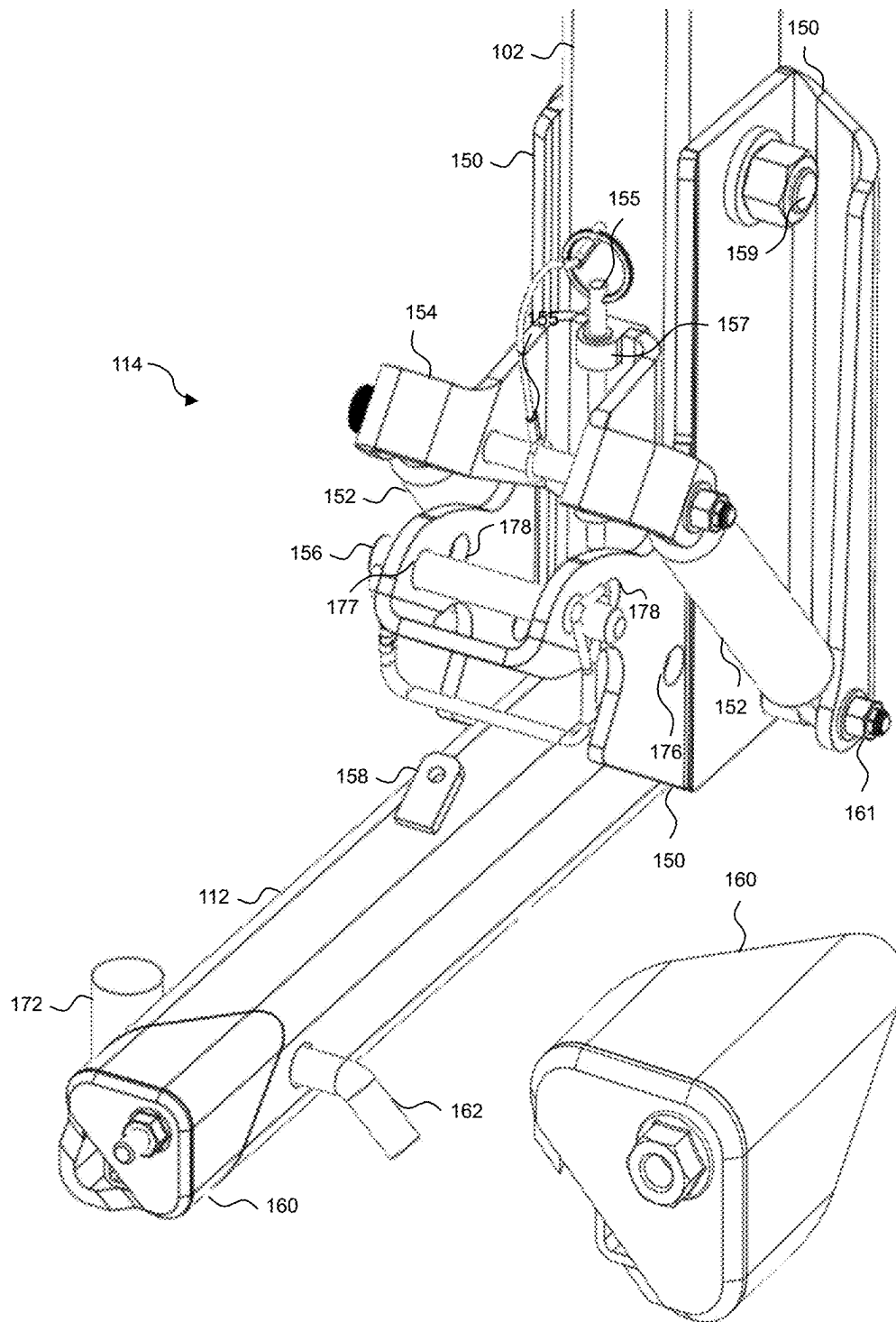


FIG. 15

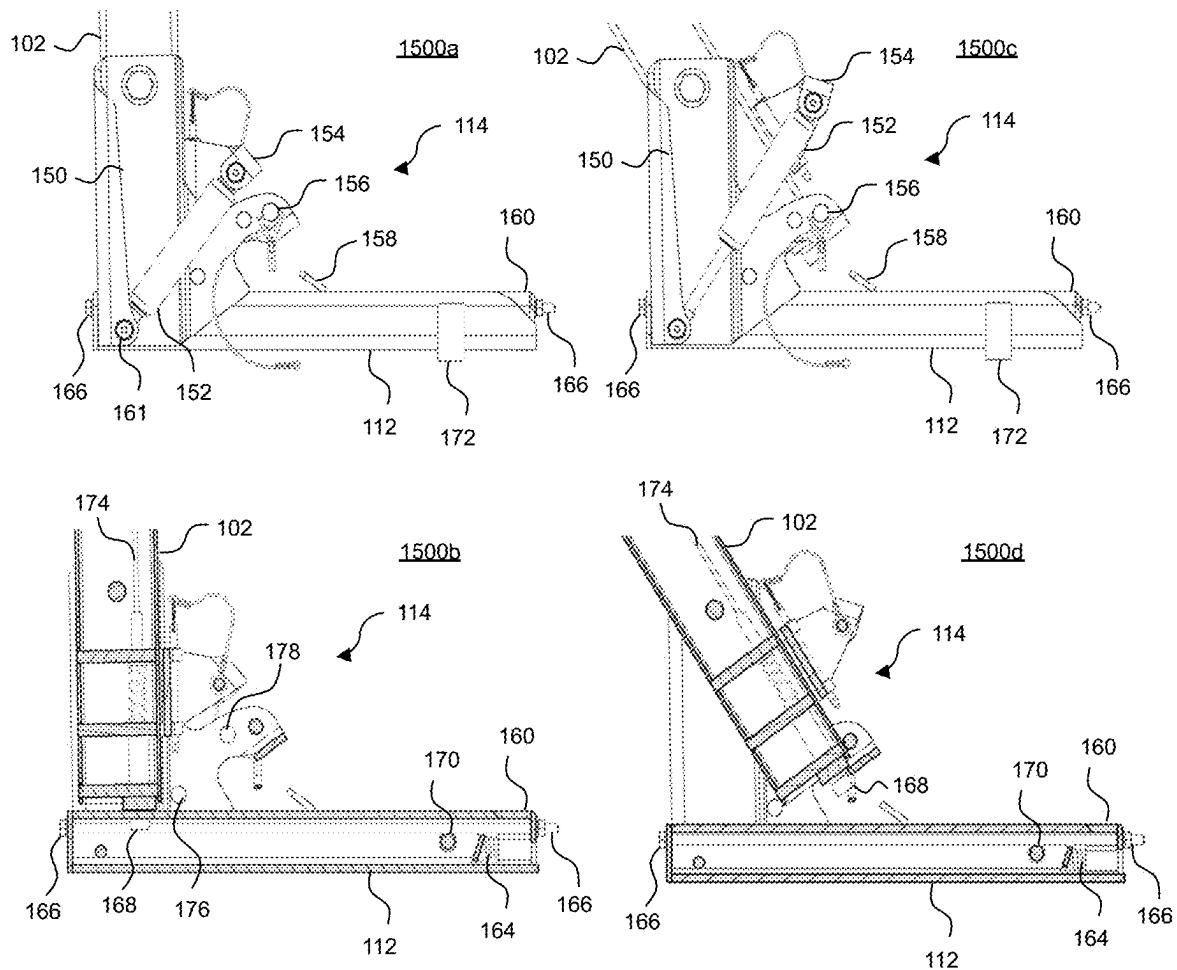


FIG. 16

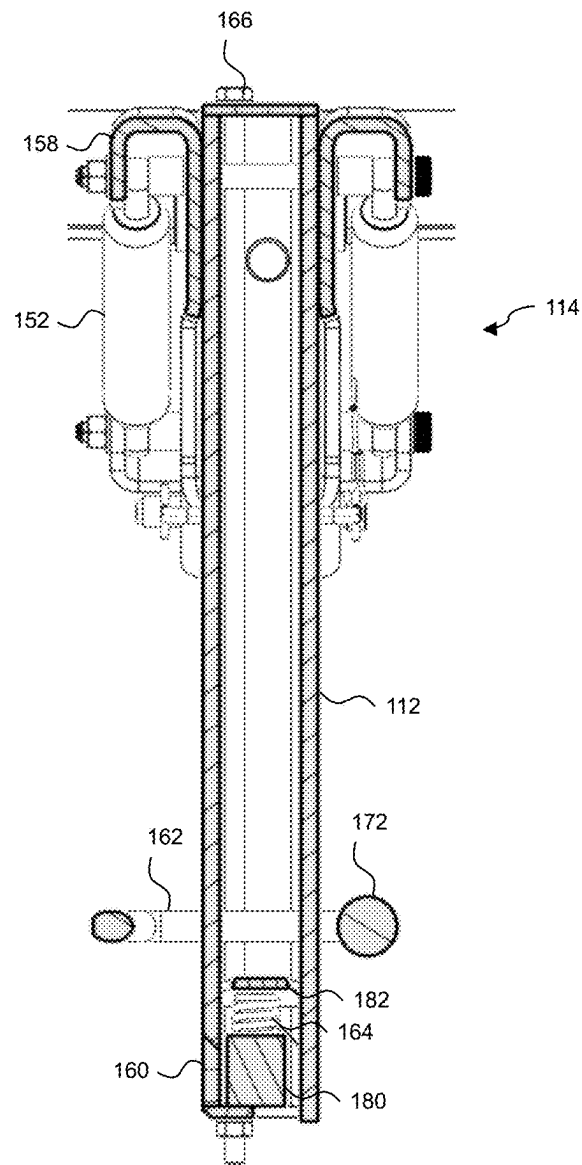


FIG. 17

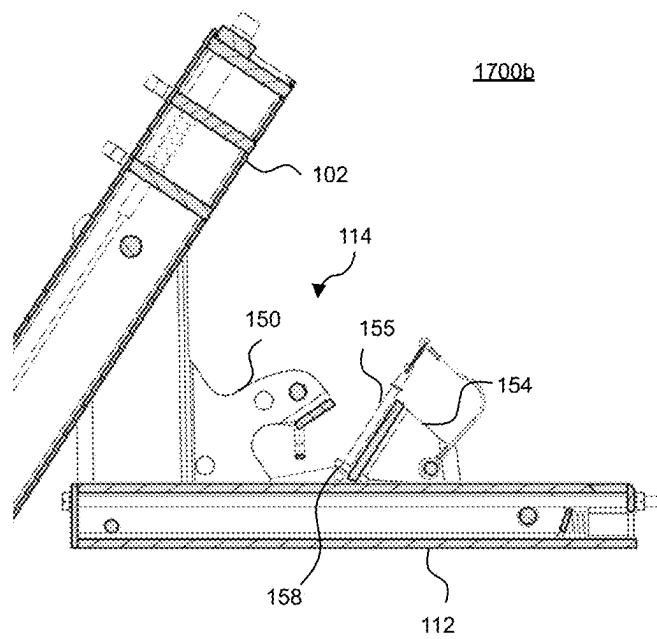
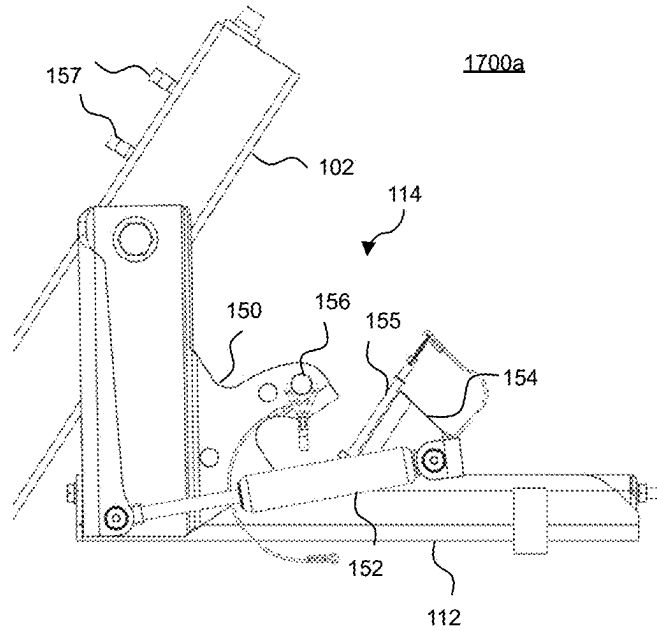


FIG. 18

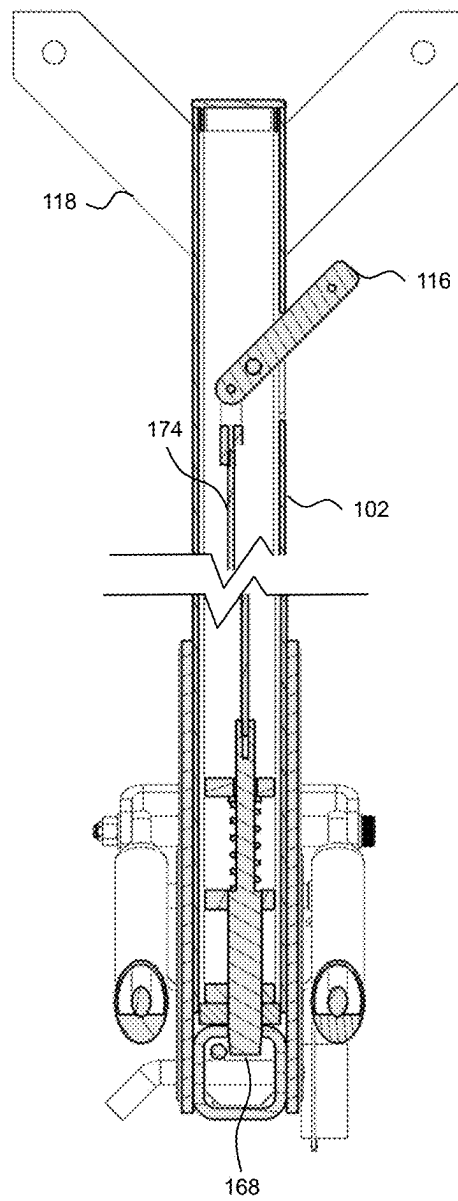


FIG. 19

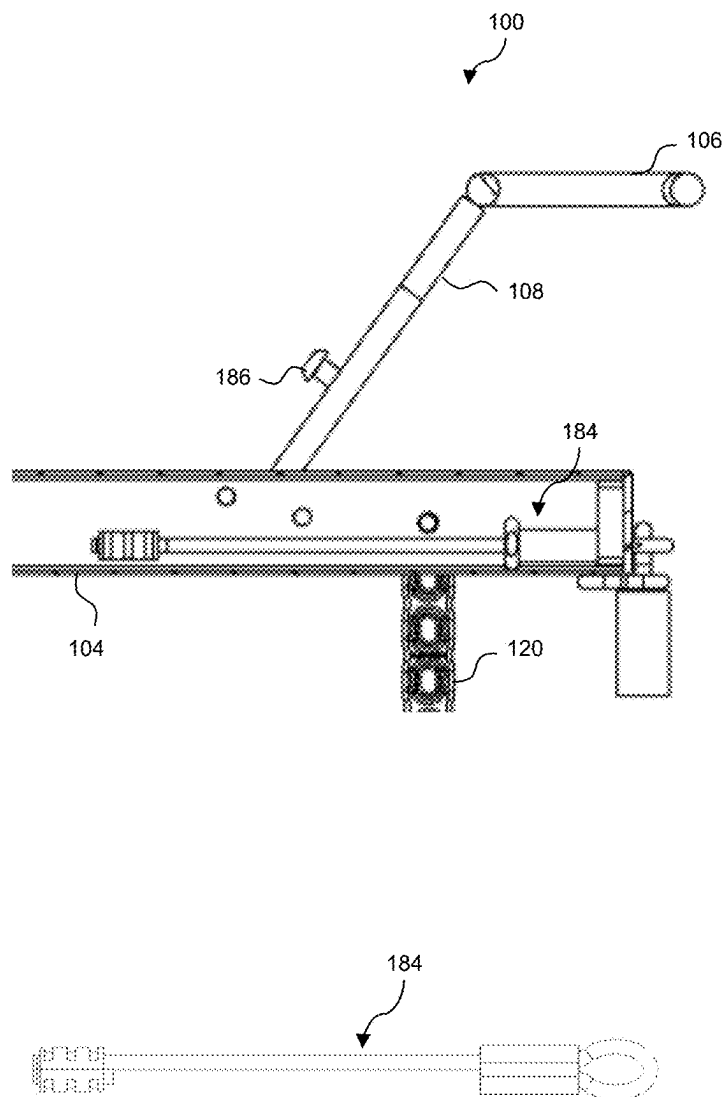


FIG. 20

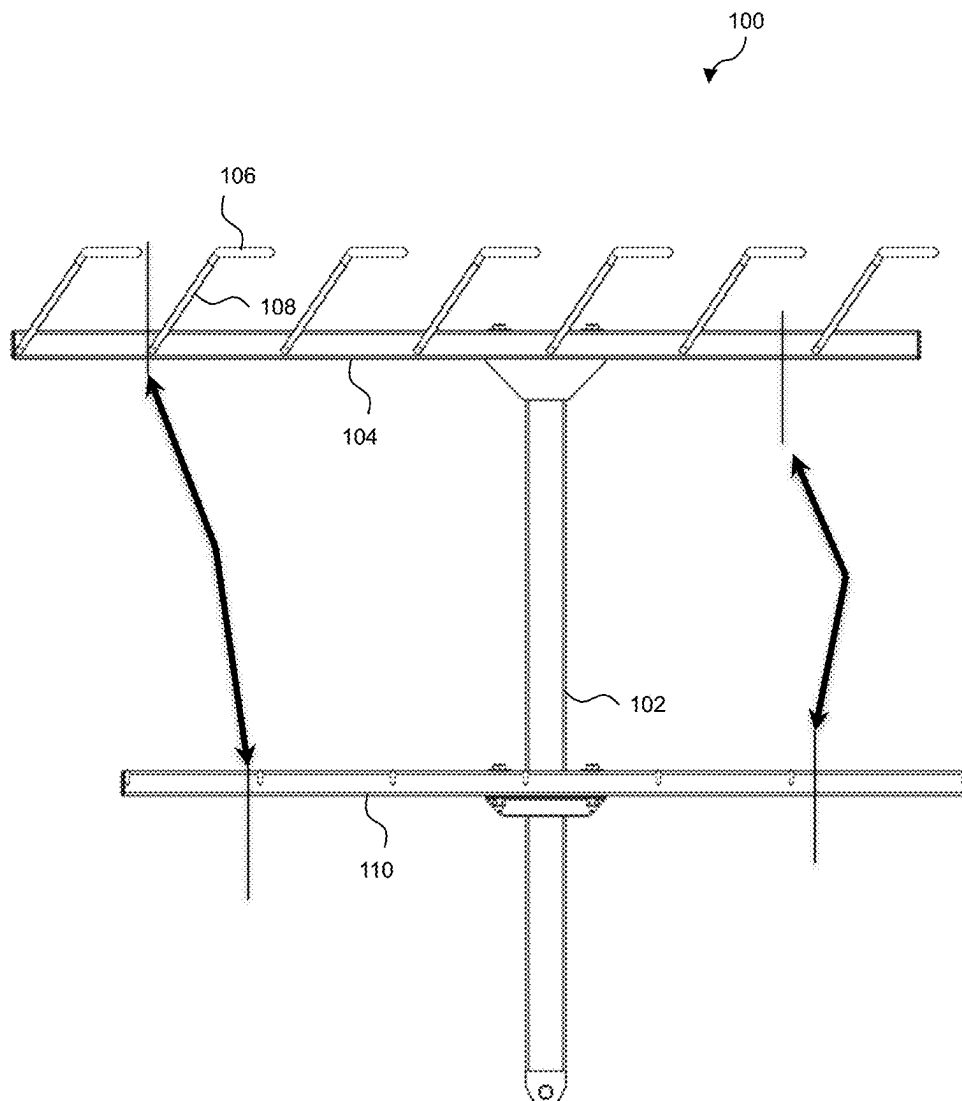


FIG. 21

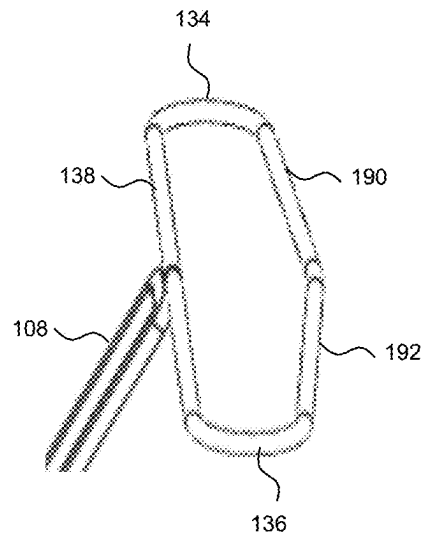


FIG. 22A

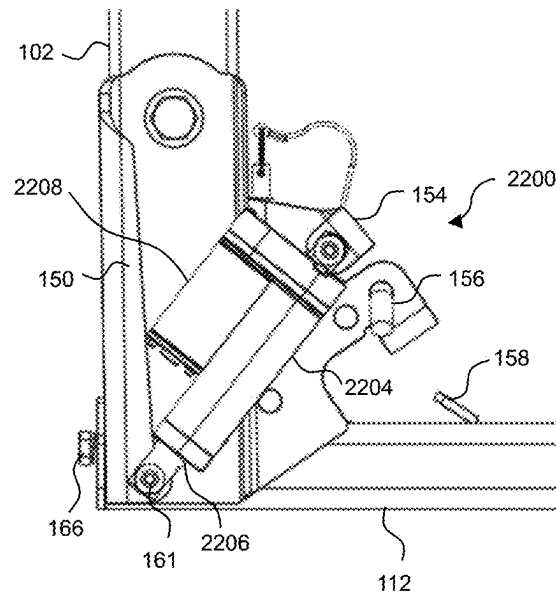


FIG. 22B

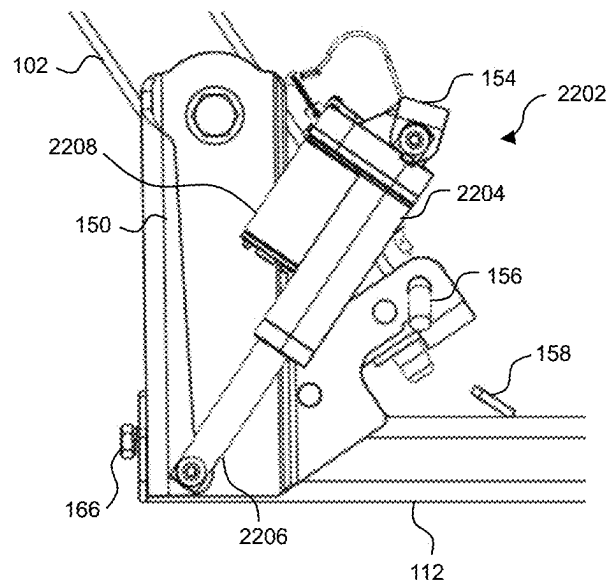
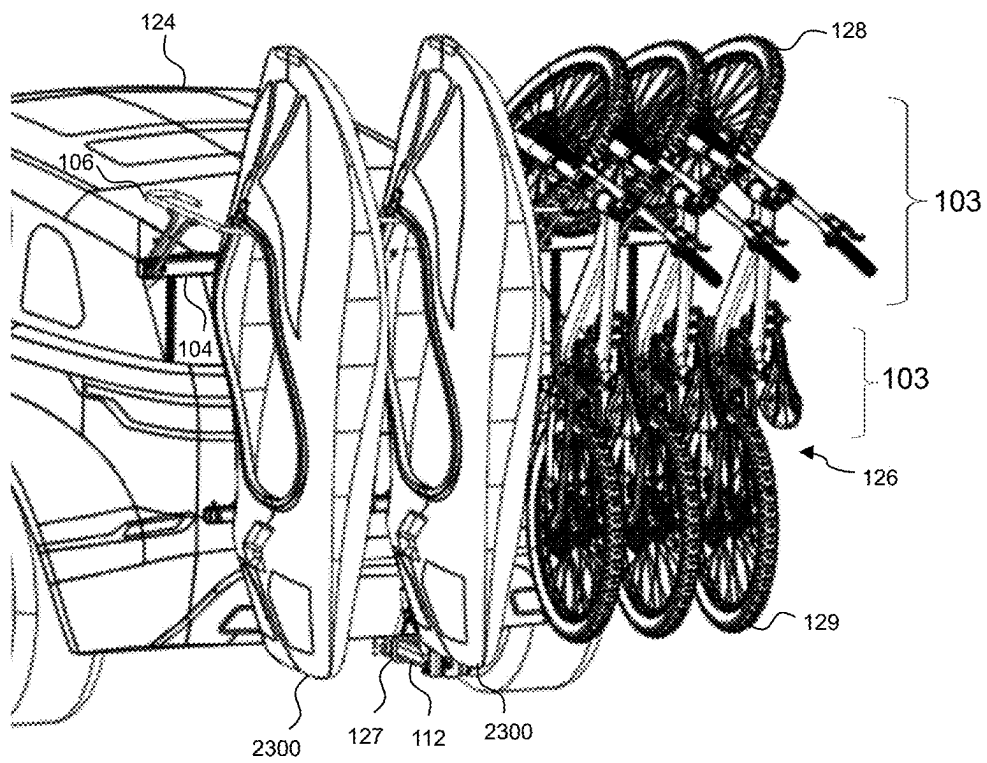


FIG. 23



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BICYCLE CARRIER WITH DAMPER**CROSS-REFERENCE TO RELATED APPLICATIONS**

The present application is a Continuation-in-Part of co-pending U.S. application Ser. No. 18/737,079, entitled "ROTATABLE HITCH-MOUNTED PAYLOAD ADAPTOR," filed Jun. 7, 2024. U.S. application Ser. No. 18/737,079 claims priority benefit from U.S. Provisional Patent Application Ser. No. 63/626,355, entitled "ROTATABLE HITCH-MOUNTED PAYLOAD ADAPTOR," filed Jan. 29, 2024. U.S. application Ser. No. 18/737,079, entitled "ROTATABLE HITCH-MOUNTED PAYLOAD ADAPTOR," is also a Continuation-in-Part of U.S. application Ser. No. 18/155,696 filed Jan. 17, 2023, now U.S. Pat. No. 12,005,866, which issued on Jun. 11, 2024. U.S. patent application Ser. No. 18/155,696 is a Continuation-in-Part of U.S. application Ser. No. 17/339,562, entitled "BICYCLE RACK WITH C-HOOPS," filed Jun. 4, 2021, which issued on Jan. 17, 2023 as U.S. Pat. No. 11,554,724. U.S. patent application Ser. No. 17/339,562 is a Continuation of U.S. application Ser. No. 17/003,094, entitled "BICYCLE CARRIER AND BICYCLE STORAGE RACK," filed Aug. 26, 2020 which issued Feb. 7, 2023 as U.S. Pat. No. 11,572,022. U.S. application Ser. No. 17/003,094 is a Continuation-in-Part of then co-pending International Patent Application No. PCT/US2019/044865 entitled "BICYCLE CARRIER AND BICYCLE STORAGE RACK," filed Aug. 2, 2019. International Patent Application No. PCT/US2019/044865 claims priority benefit from U.S. Provisional Patent Application No. 62/715,203, entitled "BICYCLE CARRIER AND BICYCLE STORAGE RACK," filed Aug. 6, 2018, now expired. International Patent Application No. PCT/US2019/044865 also claims priority benefit from U.S. Provisional Patent Application No. 62/841,933, entitled "BICYCLE CARRIER AND BICYCLE STORAGE RACK," filed May 2, 2019, now expired.

Each of the foregoing applications, to the extent not inconsistent with the disclosure herein, is incorporated by reference in its entirety.

This application is described by the extrajudicial term Continuation-in-Part to draw attention to the fact that its description differs from its immediate parent application. All terminology, structure, and function used in the description and claims is directly supported by one or more applications in its priority chain, each of which is incorporated by reference. Applicant asserts no new matter is added hereto.

SUMMARY

In an embodiment, a bicycle carrier with a hitch mount for a vehicle includes a hitch bar configured to mount to a vehicle trailer hitch receiver; a hinge assembly operatively coupled to the hitch bar, the hinge assembly including a damper operatively coupled to a support post and the hitch bar, the damper being configured to limit a rate of rotation of the support post relative to the hitch bar. The support post has a proximal end and a distal end, the support post being operatively coupled to the hinge assembly at the proximal end, the support post being rotatable relative to the hitch bar through an arc in a vertical plane defined by the hinge assembly between a lowered position and a raised position. A first support member is operatively coupled to the distal end of the support post. A wheel-receiving member is coupled to the first support member, the wheel-receiving member being configured to receive a wheel of a bicycle

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such that the bicycle is suspended vertically while the wheel is in the wheel-receiving member, the wheel-receiving member being further configured to receive the wheel at any of a range of angles with respect to the support post, the range of angles including at least an angle substantially parallel to the support post and an acute angle with respect to the support post.

BRIEF DESCRIPTION OF THE DRAWINGS

FIG. 1 is a perspective view of a bicycle carrier with a hitch mount, according to an embodiment.

FIG. 2 is an illustration of a bicycle carrier coupled to a vehicle and carrying a bicycle, according to an embodiment.

FIG. 3 illustrates a bicycle carrier carrying a bicycle and coupled to a vehicle while at a loading position, according to one embodiment.

FIG. 4 is a side view of a bicycle carrier coupled to a vehicle and in a stabilization configuration, according to an embodiment.

FIG. 5 is a side view of a bicycle carrier positioned on the ground and stabilizing a bicycle, according to an embodiment.

FIG. 6 is a view of the bicycle carrier during a loading sequence of the bicycle, according to an embodiment.

FIG. 7 is a view of the bicycle carrier after a loading sequence of the bicycle, according to an embodiment.

FIG. 8 is a view of a bicycle carrier supporting a first bicycle and a second bicycle mounted in a tune-up position, according to an embodiment.

FIG. 9A is a side view of a bicycle carrier in a wall-mounted configuration, according to an embodiment.

FIG. 9B is a view of a wall mount system used in the wall-mounted configuration of the bicycle carrier described in FIG. 9A, according to an embodiment.

FIG. 9C is a detailed view of the wall mount fittings shown in FIG. 9B, according to an embodiment.

FIG. 10 is a side view of a bicycle carrier in a loading position, according to an embodiment.

FIG. 11 is a side view of a bicycle carrier in a storage position, according to an embodiment.

FIG. 12 is a perspective view of a wheel-receiving member formed as a C-hoop and angled support members, according to an embodiment.

FIG. 13 is a perspective view of a C-hoop and angled support members, according to an alternate embodiment.

FIG. 14 is an isometric view of a hinge assembly of a bicycle carrier and wedge, according to an embodiment.

FIG. 15 illustrates four views of a hinge assembly of a bicycle carrier, according to an embodiment.

FIG. 16 is a bottom sectional view of the hinge assembly and a hitch bar, according to an embodiment.

FIG. 17 illustrates two views of a hinge assembly of a bicycle carrier, according to an embodiment.

FIG. 18 is a cross section view of a bicycle carrier post showing the lever actuation of the locking latch pin, according to an embodiment.

FIG. 19 is a side sectional view of a first horizontal support member of a bicycle carrier showing a stored internal locking cable, according to an embodiment.

FIG. 20 illustrates a bicycle carrier that can telescope to reduce the number of C-hoops, according to an embodiment.

FIG. 21 illustrates a C-hoop and a portion of angled support members, according to an embodiment.

FIG. 22A is a diagram showing a portion of a sports equipment carrier including a force transmission in a raised position, according to an embodiment.

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FIG. 22B is a diagram showing a portion of a sports equipment carrier including the force transmission of FIG. 22A in a lowered or partially lowered position, according to an embodiment.

FIG. 23 illustrates a sports equipment carrier configured to carry a combination of bicycles and whitewater kayaks, according to an embodiment.

DETAILED DESCRIPTION

In the following detailed description, reference is made to the accompanying drawings, which form a part hereof. In the drawings, similar symbols typically identify similar components, unless context dictates otherwise. Other embodiments may be used and/or other changes may be made without departing from the spirit or scope of the disclosure.

FIG. 1 is a perspective view of a bicycle carrier 100, according to an embodiment. As used herein, the terms bicycle carrier, sports equipment carrier, sports rack, and the like are used interchangeably. Unless context dictates otherwise, the terms should be considered synonymous. The bicycle carrier 100 includes a post 102 (also referenced as “support post” herein), an equipment mount 103, a hitch bar 112, and a hinge assembly 114 that rotatably couples the hitch bar 112 to the post 102. In an embodiment, the equipment mount 103 includes a first horizontal support member 104, wheel receiving members 106 including angled support members 108 configured to receive and hold bicycle wheels, and a second horizontal support member 110. The post 102 and/or equipment mount 103 may optionally include a cable lock loop 115 configured to receive one or more locking cables or chains for securing bicycles to the bicycle carrier 100.

In the view of FIG. 1, the bicycle carrier 100 is in a raised position for transport. As used herein, the terms raised position and transport position will be understood as synonymous unless context dictates otherwise. In the transport position the hinge assembly 114 holds the post 102 at a position substantially perpendicular to the hitch bar 112. The hitch bar 112 is configured to be placed in and fixed to a trailer hitch receiver of a vehicle. In the transport position, the bicycle carrier 100 can safely carry a plurality of bicycles while coupled to a moving vehicle 124.

Each wheel receiving member 106 is configured to receive a front wheel of a bicycle. The front wheel of a bicycle can be positioned, lowered into, or rolled into, a wheel receiving member 106. After being positioned in a wheel receiving member 106, the wheel of the bicycle comes to rest at an angle other than vertical. The wheel of the bicycle rests against the angled support members 108 and on the first horizontal support member 104. The bicycle is suspended vertically when the wheel is positioned in the wheel receiving member 106. This is illustrated in FIG. 2 and elsewhere in the figures. The angled support members 108 can be coupled to the first horizontal support member 104 by bolts, welds, or other coupling devices or methods.

As used herein, it will be understood that a wheel receiving member may take the form of a C-shaped hoop, also referred to as C-hoop. In description herein, unless context dictates otherwise, it will be understood that the term C-hoop may be generalized to wheel receiving member. This naming convention is supported by one or more patents and applications from which this application derives priority and incorporates by reference.

Each wheel receiving member 106 is coupled to the first horizontal support member 104 by one or more angled

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support members 108. In the example of FIG. 1, the support members 108 extend from the first horizontal support member 104 at an angle other than vertical. In one embodiment, the support members 108 extend at an angle of about 55° from horizontal. In this case, the wheel of the bicycle will rest in a wheel receiving member 106 at an angle of about 45° from horizontal. Those of skill in the art will recognize, in light of the present disclosure, that other angles are possible without departing from the scope of the present disclosure.

In one embodiment, the post 102 is about 48 inches in length. The post 102 can have width dimensions of about 2 inches by 3 inches. The first horizontal support member 104 is about 68 inches in length. The first horizontal support member 104 has width dimensions of about 2 inches by 2 inches. The second horizontal support member 110 has a length of about 64 inches. The second horizontal support member 110 has width dimensions of about 1.75 inches by 1.75 inches. The hitch bar 112 has a length of about 20 inches, with width dimensions of about 2 inches by 2 inches. In one embodiment, the post 102, the first horizontal support member 104, the second horizontal support member 110, and the hitch bar 112 are each made of steel. Those of skill in the art will recognize, in light of the present disclosure, that dimensions and materials described herein are given by way of example and that other suitable dimensions and materials can be utilized without departing from the scope of the present disclosure.

In one embodiment, the first horizontal support member 104 is coupled to the post 102 by mounting tabs 118. The mounting tabs 118 can include steel and can be fixed to the first horizontal support member 104 by bolts, by welding, or by other fasteners or methods. The mounting tabs 118 can be fixed to the post 102 by welding, bolts, or by other fasteners or methods.

In one embodiment, the post 102 includes a release lever 116. Pulling the release lever 116 enables the post 102 to be rotated relative to the hitch bar 112 via the hinge assembly 114. The post 102 can be rotated between various stopping positions, as will be described in greater detail below. The structure, components, and function of the hinge assembly 114 will also be described in greater detail below with respect to FIGS. 14-17.

In one embodiment, the bicycle carrier 100 includes a plurality of first straps 120 each coupled to the first horizontal support member 104. There is a first strap 120 for each wheel receiving member 106. When a bicycle wheel is placed in a wheel receiving member 106, the corresponding first strap 120 is looped around the wheel between two of the spokes and is fastened to a strap anchor on one of the support members 108. This can keep the wheel from moving or shifting when positioned in the wheel receiving member 106.

In one embodiment, the bicycle carrier 100 includes a plurality of second straps 122 each coupled to the second horizontal support member 110. There is a second strap 122 for each wheel receiving member 106. When a bicycle is suspended from a wheel receiving member 106 by placing the wheel in the wheel receiving member 106, the bicycle will be suspended vertically, and the rear wheel will be in contact with the second horizontal support member 110. The second strap 122 can be looped between two spokes on the rear wheel and connected to a fastener positioned on the second horizontal support member 110. This will keep the rear wheel from shifting relative to the second horizontal support member 110 during transport. Those of skill in the art will recognize, in light of the present

disclosure, that other fastening methods can be used to secure the bicycle suspended from one of the wheel receiving members 106.

The word vertical in the term post 102 refers to the position of the post 102 when in the transport configuration. The transport configuration corresponds to the configuration in which the bicycle carrier 100 is coupled to a vehicle, is carrying one or more bicycles 126, 126a, 126b, and is actively being transported. In this case, the post 102 is substantially perpendicular to the hitch bar 112 and extends substantially vertically relative to flat ground. Those of skill in the art will recognize, that the post 102 can be rotated to positions other than vertical. Additionally, the bicycle carrier 100 can be laid in positions in which the post 102 would not extend in a vertical direction. Accordingly, the word vertical in the term post 102 refers to the orientation of the post 102 in a particular transport configuration. Similarly, the word horizontal in the terms first and second horizontal support member 104, 110, refers to the orientation of the support member 104, 110 during the transport configuration. The bicycle carrier 100 could be laid in positions in which the first and the second horizontal support members 104, 110 would not extend in the horizontal direction. In one embodiment, the bicycle carrier 100 can include a post 102 that is not necessarily vertical in the transporting position. Additionally, the bicycle carrier 100 can include first and second support members 104, 110 that are not horizontal.

FIG. 2 is an illustration of a bicycle carrier 100 coupled to a vehicle 124 and carrying a bicycle 126, according to an embodiment. The hitch bar 112 of the bicycle carrier 100 has been inserted in and coupled to a trailer hitch receiver 127 of the vehicle 124. The hitch bar 112 can be coupled to the trailer hitch receiver 127 by bolts and pins or other common fastening configurations, as will be described in more detail below.

The front wheel 128 of the bicycle 126 is positioned in one of the wheel receiving members 106, here depicted as a C-hoop. When the front wheel 128 is positioned in the C-hoop 106, the front wheel 128 rests at a non-vertical angle. The front wheel 128 rests on the first horizontal support member 104. The front wheel 128 is also in contact with one or more support members 108. The bicycle 126 is suspended vertically. The rear wheel 129 rests against the second horizontal support member 110. When describing how the front and the rear wheels 128, 129 rest on or are supported by various support members, it is understood that the tires may actually be in contact with the various support members.

The post 102 is in the transport position in which the post 102 extends substantially perpendicularly from the hitch bar 112.

While FIG. 2 shows a single bicycle 126 carried by the bicycle carrier 100, in practice, the bicycle carrier 100 can carry multiple bicycles 126. In particular, the bicycle carrier 100 can carry a bicycle 126 for each C-hoop 106. Accordingly, each C-hoop 106 can receive a front wheel 128 of a respective bicycle 126.

FIG. 3 illustrates a bicycle carrier 100 carrying a bicycle 126 and coupled to a vehicle 124 while at a loading position, according to one embodiment. The hinge assembly 114 defines a loading stop position. When the release lever 116 is pulled, the hinge assembly 114 enables the post 102 to rotate relative to the hinge bar 112 to a loading stop position. At the loading stop position, the post 102 is oriented at about 33° from vertical, in one example. This enables bicycles 126 to more easily be loaded onto or off of the bicycle carrier 100. This is because the C-hoops 106 are both lower to the

ground and oriented at an angle that facilitates easy loading and unloading of bicycles 126.

As will be described in more detail below, hinge motion is slowed by dampers/gas springs included in the hinge assembly 114. This damping enhances the safety of rotating between the transport position and the load position.

In one embodiment, each bicycle 126 can be accessed without the removal of other bicycles 126. Loading bicycles 126 can also be done in the full vertical transport mode or in the angled back loading position. In one embodiment, moving from the hinged position requires an upward and forward force toward the vehicle 124 to move from the loading (hinged) position until the post 102 automatically locks in the vertical or transport position.

FIG. 4 is a side view of a bicycle carrier 100 coupled to a vehicle 124 and in a stabilization configuration, according to an embodiment. In the stabilization configuration, the post 102 is rotated more than 90° from the transport position by operation of the hinge assembly 114. The C-hoops 106 rest on the ground. A bicycle 126 is stabilized by the bicycle carrier 100. In particular, the rear wheel 129 of the bicycle 126 is positioned in the C-hoop 106. The C-hoop 106 stabilizes the rear wheel 129 so that the bicycle 126 will not fall over. The first strap 120 can be utilized to strap the rear wheel 129 to the C-hoop 106.

This configuration is useful in situations in which the vehicle 124 will be parked and bicycles 126 will be periodically used. When a bicycle 126 is not used, it can be positioned in the C-hoop 106 as shown in FIG. 4. When a bicycle 126 is to be used, the bicycle 126 can easily be removed from the C-hoop 106. This configuration can be particularly useful in situations like camping when bicycles 126 may be ridden to and from camp. Instead of being laid on the ground are rested against trees or other structures, each bicycle 126 can be positioned in a C-hoop 106 for stabilization.

FIG. 5 is a side view of a bicycle carrier 100 positioned on the ground and stabilizing a bicycle 126, according to an embodiment. In this example, the post 102 and the hinge part 112 are oriented in the transport position by the hinge assembly 114. However, the bicycle carrier 100 is not coupled to a vehicle 124. Instead, the bicycle carrier 100 is laid on the ground. The C-hoops 106 rest on the ground similar to the configuration shown in FIG. 4. The bicycle carrier 100 can stabilize a plurality of bicycles 126 in this configuration by positioning the rear wheels 129 of the bicycles 126 in the C-hoops 106.

FIG. 6 is a view of the bicycle carrier 100 during a loading sequence of the bicycle 126, according to an embodiment. The front wheel 128 of the bicycle 126 is lowered into the C-hoop 106. While the front wheel 128 is loaded into the C-hoop 106, the front wheel 128 can be substantially vertically oriented. After loading, the front wheel 128 will settle at an angle other than vertical.

FIG. 7 is a view of the bicycle carrier 100 after a loading sequence of the bicycle 126, according to an embodiment. The front wheel 128 of the bicycle 126 is positioned in the C-hoop 106. The front wheel 128 has settled to a position such that the front wheel 128 is not oriented vertically, but rather is oriented at an angle relative to vertical. In one example, the angle is 45°.

FIG. 8 is a view of a bicycle carrier 100 supporting a first bicycle 126a and a second bicycle 126b mounted in a tune-up position, according to an embodiment. The first bicycle 126a is supported in a standard fashion with the front wheel 128 positioned in a C-hoop 106. The second bicycle 126b is supported by positioning a seat post within a C-hoop

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106 such that a seat **125** of the second bicycle **126b** rests on and is supported by the C-hoops **106**.

The shape of the C-hoop **106** enables the second bicycle **126b** to be suspended by the seat **125**. Because the C-hoop **106** includes first and second terminations **130**, **132** (see FIGS. **12** and **13**), the seat post of the second bicycle **126b** can be easily maneuvered through a gap **131** (see FIGS. **12** and **13**). The second bicycle **126b** can then be suspended by the seat **125**, as shown in FIG. **8**.

In one embodiment, when the second bicycle **126b** is suspended by the seat **125**, maintenance or repairs can be performed easily on the second bicycle **126b** because the wheel and pedals are free to rotate. Accordingly, the bicycle carrier **100** can act as a repair station for bicycles **126**.

FIG. **9A** is a side view of a bicycle carrier **100** in a wall-mounted configuration, according to an embodiment. The hinge assembly **114** is operated such that the post **102** has rotated more than 90° from the transport position to a storage or wall-mounted position. In one example, in the wall-mounted or storage position, the post **102** is rotated about 150° relative to the hinge bar **112** from the transport position.

In one embodiment, the wall-mounted position is suitable for coupling the bicycle carrier **100** to a wall mount structure **144**. The wall mount structure **144** can include one or more boards, beams, or bent sheet metal coupled to a wall **145** of a structure, such as a garage. One or more hooks **146** are coupled to the wall mount structure **144**. The bicycle carrier **100** is coupled to the wall mount structure **144** by positioning one or more of the C-hoops **106** on one or more of the hooks **146**. The post **102** and the hitch bar **112** are in contact with a floor **148** or on top of a small spacer sitting on the floor **148**.

In the wall-mounted position, a bicycle **126** can be loaded into a C-hoop **106** by rolling the back wheel **129** with the front wheel **128** in the air until the front wheel **128** can be positioned in the C-hoop **106**. In some cases, this may require lifting the bicycle **126** to enable the front wheel **128** to slide into the C-hoop **106**. The downward tilting angle of the C-hoop **106** makes it relatively easy to place the front wheel **128** of a bicycle **126** in the C-hoop **106**. Each C-hoop **106** of the bicycle carrier **100** can receive a respective bicycle **126** in the wall-mounted configuration.

FIG. **9B** is a view of a wall mount system used in the wall-mounted configuration of the bicycle carrier **100** described in FIG. **9A**, according to an embodiment. FIG. **9C** is a detailed view of the wall mount fittings **149** described in FIG. **9B**, according to an embodiment.

Referring to FIGS. **9B** and **9C**, in an embodiment, the bicycle carrier **100** further includes a wall mount bar **144** (e.g., wall mount structure **144** of FIG. **9A**) configured to be screwed to a wall **145**, such as using wood screws to be screwed into wall studs, and one or more wall mount fittings **149** (such as one or more hooks **146**) configured to be coupled to the wall mount bar **144** and configured to receive and hold a feature of the equipment mount **103**. In an embodiment, the wall mount system uses two wall mount fittings **149**, the two wall mount fittings **149** being configured to hook around respective C-hoops **106**.

FIG. **10** is a side view of a bicycle carrier **100** in a loading position, according to an embodiment. The hinge assembly **114** has enabled the post **102** to rotate about 33 degrees from the transport position to the loading position.

FIG. **11** is a side view of a bicycle carrier **100** in a storage position, according to an embodiment. The hinge assembly **114** has enabled the post **102** to rotate about 150 degrees from the transport position to the storage position.

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FIG. **12** is a perspective view of a C-hoop **106** and angled support members **108**, according to an embodiment. The C-hoop **106** is substantially in the shape of a letter C. The C-hoop **106** includes two terminations, **130**, **132**. The two terminations **130**, **132** define a gap **131** between the two terminations **130**, **132**. The C-hoop **106** includes a first bend **134**, a second bend **136**, and a straight portion **138** extending between the first and the second bends **134**, **136**.

In one embodiment, the length of the C-hoop **106** is 26 inches. The length is defined as the internal distance between the farthest points of the first bend **134** and the second bend **136**. Those of skill in the art will recognize, in light of the present disclosure, that a C-hoop **106** can have a length other than 26 inches without departing from the scope of the present disclosure.

In one embodiment, the width of the C-hoop **106** is 3.5 inches. The width is defined as the shortest distance between the straight portion **138** and either of the terminations **130** or **132**. Those of skill in the art will recognize, in light of the present disclosure, that a C-hoop **106** can have a width other than 3.5 inches without departing from the scope of the present disclosure.

In one embodiment, the distance D between the first termination **130** and the second termination **132** is about 10 inches. Accordingly, the gap **131** between the first and the second terminations **130**, **132** is 10 inches. Those of skill in the art will recognize, in light of the present disclosure, that the C-hoop **106** can include a gap **131** of other than 10 inches between the first and the second terminations **130**, **132** without departing from the scope of the present disclosure.

In one embodiment, the shape of the C-hoop **106** provides many benefits. For example, when a front wheel **128** is inserted into the C-hoop **106**, the front wheel **128** can enter at an angle substantially perpendicular to the plane defined by the C-hoop **106**. Because the C-hoop **106** terminates at the terminations **130**, **132** without additional structure extending downward from the area of the terminations **130**, **132**, the front wheel **128** can be loaded at a convenient perpendicular angle. If the C-hoop **106** did not terminate at the terminations **130**, **132**, but rather extended downward at an angle similar to the angled support members **108**, then the front wheel **128** would need to be loaded at an angle other than perpendicular so it can slide into the C-hoop **106**. Loading the front wheel **128** at an angle other than perpendicular is highly inconvenient and awkward. Accordingly, the shape of the C-hoop **106** as shown in FIG. **12** avoids this drawback. After the front wheel **128** has been inserted into the C-hoop **106**, the front wheel **128** will then come to a resting position at an angle other than perpendicular. In the resting position, the tire is in contact with an interior surface of the angled support members **108** and the first horizontal support member **104**.

Another benefit of the C-hoop **106** is that the gap **131** is situated so that in the resting position, the spokes of the front wheel **128** do not contact the C-hoop **106**. If the C-hoop **106** was a complete hoop instead of having the gap **131**, then it is possible that the spokes would be in contact with the C-hoop **106** when in the resting position. The resulting pressure on the spokes could result in bent or otherwise damaged spokes.

In one embodiment, the C-hoop **106** is a single integral structure. For example, the C-hoop **106** can correspond to a single bar or rod that is bent into the shape shown in FIG. **12**. Alternatively, the C-hoop **106** can include multiple sections connected together to form the shape shown in FIG. **12**.

In one embodiment, the C-hoop **106** is steel. Other sufficiently strong materials can be used for the C-hoop **106**.

In one embodiment, when the bicycle carrier **100** is in the transport position, the C-hoop **106** defines a plane substantially parallel to the ground, or perpendicular to the vertical direction. For example, if a flat board is laid on top of the C-hoop **106** when in the transporting configuration, the plane of the board would be parallel to the ground or perpendicular to the vertical direction.

FIG. **13** is a perspective view of a C-hoop **106** and angled support members **108**, according to an alternate embodiment. The C-hoop **106** is substantially in the shape of a C. However, instead of a straight portion **138** extending between the first and the second bends **134**, **136**, a bent portion **142** extends between the first and the second bends **134**, **136**. Otherwise, the C-hoop **106** of FIG. **13** is substantially similar to and provides similar benefits as the C-hoop **106** of FIG. **12**.

FIGS. **14-16** include illustrations of a hitch bar wedge clamp, according to an embodiment.

FIG. **14** is an isometric view of a hinge assembly **114** of a bicycle carrier **100** and wedge **160**, according to an embodiment. FIG. **14** also shows portions of the hitch bar **112** and post **102** coupled to the hinge assembly **114**. The hinge assembly **114** allows rotation of the post **102** relative to the hitch bar **112**.

In one embodiment, the hinge assembly **114** includes a bracket **150**, dampers **152**, a damper yoke **154**, a damper pin **155**, a hinge stop pin **156**, and a bolt **159**. The bracket **150** is fixed to the hitch bar **112** by one or more bolts or welds. The bolt **159** is coupled to the bracket **150** and passes through the post **102**. The bolt **159** acts as an axle for rotation of the post **102**. When the post **102** rotates, it rotates about the bolt **159**.

In one embodiment, the bracket **150** is a unitary structure. In one embodiment, the bracket **150** is steel.

In one embodiment, when the release lever **116** is pulled, the hinge assembly **114** enables rotation of the post **102** relative to the hitch bar **112**. The dampers **152** slow, resist, or oppose the rotation. This enhances the safety of the bicycle carrier **100**. This is because when rotating between the transport and the loading positions, the rotation cannot happen rapidly due to the dampers **152**. This ensures that the post **102** will not quickly fall or rotate when rotation is enabled, potentially injuring somebody or damaging the bicycle carrier **100**. Instead, after rotation is enabled by pulling the release lever **116**, some force may need to be applied by an individual to cause rotation against the force of the dampers **152** when no bicycles are on the rack.

In one embodiment, the dampers **152** include hydraulic damping. The dampers **152** can include the fluid chamber and a piston. Flow of a fluid into or out of the fluid chamber is inhibited by a relatively small aperture, and potentially by the viscosity of the fluid. Rotation of the post **102** relative to the hitch bar **112** is opposed by this hydraulic effect. In one embodiment, the fluid is a liquid. In one embodiment, the fluid is an oil. In one embodiment, the fluid is a gas.

In one embodiment, the dampers **152** can include one or more springs that resist rotation of the post **102** relative to the hitch bar **112**. In one embodiment, the dampers **152** can include gas springs. Those of skill in the art will recognize, in light of the present disclosure, that many other types of dampers **152** can be used without departing from the scope of the present disclosure.

In one embodiment, the dampers **152** are mounted between the hitch bar **112** and the damper yoke **154**. The dampers **152** are coupled to the hitch bar **112** by a bolt **161**.

In one embodiment, the hinge stop pin **156** defines a stop point for rotation of the post **102** relative to the hitch bar **112**. In particular, when rotating from the transport position to the loading position, the hinge stop pin **156** stops rotation at the transport position. In one embodiment, the loading position is a 33° rotation from the transport position. The bracket **150** includes multiple sets of apertures **176**, **177**, **178** for receiving the hinge stop pin **156**. The hinge stop pin **156** is placed in the apertures **177** in FIG. **14**. The hinge stop pin **156** can be placed in the apertures **178** to reduce the angle of the loading position relative to the transport position. The hinge stop pin **156** can be placed in the apertures **176** to substantially prohibit rotation of the post **102** from the transport position. The hinge stop pin **156** can be tied to the bracket **150** by a tie.

The post **102** includes first damper pin receivers **157** for receiving a damper pin **155**. The damper yoke **154** is placed in the damping position as shown in FIG. **14**. In the damping position, the dampers **152** dampen or resist rotation of the post **102**. The damper pin **155** is placed in the first damper pin receivers **157**, thereby fixing the damper yoke **154** to the post **102** in the damping position. The damper pin **155** can be tied to the damper yoke **154** by a tie.

In one embodiment, to enable rotation of the post **102** to the wall mounting position, the damper pin **155** and the hinge stop pin **156** are removed. With the damper pin **155** removed, the damper yoke **154** can be decoupled from the post **102** and moved to a damper release position in which the dampers **152** do not dampen or resist rotation of the post **102**. With the damper pin **155** removed from the first damper pin receivers **157**, the hinge stop pin **156** removed, and the release lever **116** pulled, the post **102** can rotate beyond the loading position to the wall mounting position. This can correspond to rotation of up to 150 degrees from the transport position.

The hitch bar **112** includes a second damper pin receiver **158**. The damper pin **155** can be received in the second damper pin receiver **158** to secure the damper yoke **154** in the damper release position.

In one embodiment, a hitch wedge **160** helps to secure the hitch bar **112** in the hitch receiver **127** (e.g., trailer hitch receiver **127**) of a vehicle **124**. For example, after the hitch bar **112** is positioned in the hitch receiver **127**, the hitch wedge **160** will be positioned within the hitch receiver **127**. A hitch pin **162** can then be placed through apertures in the hitch receiver **127** and apertures **170** (see FIG. **15**) in the hitch bar **112**. A hitch pin lock **172** can lock the hitch pin **162** in place. A hitch wedge bolt **166** (see FIG. **15**) can then be tightened. The tightening of the hitch wedge bolt **166** pulls or tightens the hitch wedge **160** into place, thus locking the entire bicycle carrier **100** to the vehicle hitch mounting point and stopping relative motion between the bicycle carrier **100** and the vehicle **124**. This greatly reduces rattling of the bicycle carrier **100** and, by extension, the bicycles **126**, during transport. Loosening the hitch wedge bolt **166** loosens the hitch wedge **160**, allowing a hitch wedge spring **164** (see FIG. **15**) to fully release the hitch wedge **160**, thereby allowing the bicycle carrier **100** to be removed from the vehicle **124**. FIG. **14** also illustrates an expanded view of the hitch wedge **160**.

In one embodiment, as the hitch wedge **160** is tightened, the hitch wedge **160** presses against an interior of the hitch receiver **127** of the vehicle **124**, preventing motion of the hitch bar **112** relative to the hitch receiver **127** of the vehicle **124**. This effect can be sufficient to secure the bicycle carrier **100** to the vehicle **124** without using the hitch pin **162**.

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In one embodiment, the hinge assembly 114 also includes a damper pin 155. By pulling the damper pin 155, the dampers 152 are decoupled from the post 102. The damper yoke 154 and the dampers 152 can be rotated downward and the damper pin 155 can be placed in the second damper pin receiver 158. In this configuration, the post 102 can rotate beyond the loading position into the wall mounting or storage position.

FIG. 15 illustrates four views 1500a, 1500b, 1500c, 1500d, of a hinge assembly 114 of a bicycle carrier 100, according to an embodiment. View 1500a is a side view of a portion of the hinge assembly 114, the hitch bar 112, and the portion of the post 102 in the transport position. View 1500b is a side sectional view of the hinge assembly 114, the hitch bar 112, and a portion of the post 102 in the transport position. View 1500c is a side view of the hinge assembly 114, the hitch bar 112, and a portion of the post 102 in the loading position. View 1500d is a side sectional view of the hinge assembly 114, the hitch bar 112, and a portion of the post 102 in the loading position.

In view 1500a, the post 102 is in the transport position. The post 102 is substantially perpendicular to the hitch bar 112. The dampers 152 are in a retracted position.

In view 1500b, the post 102 is in the transport position. A release rod 174 extends through the post 102. The release rod 174 can be a release cable. When the release lever 116 is pulled, a locking latch pin 168 is released by action of the release rod 174. This enables rotation of the post 102. The side sectional view of 1500b also illustrates the hitch wedge bolt 166 extending through the hitch bar 112 to the hitch wedge 160. The side sectional view of 1500b also illustrates apertures 170 in the hitch bar 112, through which the hitch pin 162 can be passed.

In view 1500c, the post 102 is in the loading position. The locking latch pin 168 has been released, enabling rotation of the post 102 to the loading position. The loading position is achieved when the post 102 abuts the hinge stop pin 156. The hinge stop pin 156 can be placed in the aperture 178 (see view 1500b) to reduce the rotation angle of the post 102 relative to vertical. If the hinge stop pin 156 is placed in the aperture 176 (see view 1500b), rotation of the post 102 from the transport position is prohibited. This locks the post 102 in the transport position. In the view 1500d, the interior of the post 102 can be seen at the loading position.

FIG. 16 is a bottom sectional view of the hinge assembly 114 and the hitch bar 112, including a wedge 160, according to an embodiment. From the bottom sectional view, a wedge spring bracket 180, the hitch wedge spring 164, and a wedge spring stop 182 are apparent.

In one embodiment, the hitch bar 112 extends full length, greatly simplifying the insertion process because the load of the rack can be fully transferred to the hitch bar 112 as it is inserted into the hitch receiver 127. The hitch bar 112 may be formed with one upper corner and about 1/4 of the top wall and side wall abutting the upper corner extending the full length. The wedge 160 is formed to nominally occupy a space sloping away from the one upper corner, such that tightening the hitch wedge bolt 166 causes the wedge to slide along the sloping end of the hitch bar 112 to expand and securely hold the hitch bar in place. The internal spring 164 substantially prevents the wedge 160 from engaging too early in the insertion process, allowing it to be inserted easily. The orientation of the wedge spring bracket 180, wedge spring stop 182 and spring 164 in the wedge 160 also provides pressure on the wedge 160 to hold it in place that keeps it lined up with the hitch bar 112 for easier insertion.

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FIG. 17 illustrates two views 1700a and 1700b of a hinge assembly 114 of a bicycle carrier 100, according to an embodiment. View 1700a is a side view of the hinge assembly 114, the hitch bar 112, and a portion of the post 102 in the wall mounting or storage position. View 1700b is a side sectional view of the hinge assembly 114, the hitch bar 112 and a portion of the post 102 in the transport position. To enter into the storage or the mounting position, the damper pin 155 is removed, the damper yoke 154 is removed and lowered to the hitch bar 112 and the damper pin 155 is placed in the second damper pin receiver 158. With the dampers 152 and the damper yoke 154 decoupled from the post 102, and the hinge stop pin 156 removed, the post 102 can rotate beyond the loading position into the storage or wall mounting position. In the storage or wall mounting position, the post 102 is rotated more than 90 degrees from the transport position, for example 150 degrees.

FIG. 18 is a cross section view of a bicycle carrier post 102 showing the lever actuation of the locking latch pin 168. The rack release lever 116 is moved downward causing the locking latch pin 168 to release the post 102 to rotate to one of the stopping positions, such as the transport position, the loading position, or the mounting or storage position.

FIG. 19 is a side sectional view of a first horizontal support member 104 of a bicycle carrier 100, according to an embodiment. The view of FIG. 19 illustrates that the first horizontal support member 104 is hollow. An internal locking cable 184 is positioned within the hollow first horizontal support member 104. An end of the first horizontal support member 104 can be opened and the internal locking cable 184 can be drawn out. The internal locking cable 184 can then be used to lock any or all of the bicycles 126 that are carried by the bicycle carrier 100. FIG. 19 also illustrates a strap anchor 186 coupled to the support member 108. The first strap 120 can be looped between the spokes of a front wheel 128 and then latched onto the strap anchor 186.

FIG. 20 illustrates a bicycle carrier 100 that can telescope to reduce the number of C-hoops 106, according to an embodiment. The first and the second horizontal support members 104, 110 can telescope between maximum lengths and minimum lengths. For example, after removing the two C-hoops 106 on either end of the first horizontal support member 104 via removing screws or bolts, portions of the first horizontal support member 104 can be removed from or telescoped into an inferior of the first horizontal support member 104. The second horizontal support member 110 can be removed or telescoped in the same manner. In one example, the bicycle carrier 100 includes a maximum number of seven C-hoops 106. After removal and telescoping, the bicycle carrier 100 includes a minimum of five C-hoops 106. Those of skill in the art will recognize in light of the present disclosure that other maximum or minimum numbers of C-hoops 106 are possible without departing from the scope of the present disclosure.

FIG. 21 illustrates a C-hoop 106 and a portion of angled support members 108, in accordance with an embodiment. The C-hoop 106 includes a straight portion 138 coupled to the angled support members 108 extending between a first bend 134 and a second bend 136. The C-hoop 106 includes lengths 190 and 192 extending from the first and the second bends 134 and 136. The lengths 190 and 192 extend outward. This outward extension helps to ensure that the spokes of a bicycle wheel will not contact the lengths 190 and 192 when the wheel is positioned in the C-hoop 106. The lengths 190 and 192 may connect to each other. Alternatively, the lengths 190, 192 can end in terminations 130, 132 that define

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a gap **131**, as shown in FIGS. **12** and **13**. In one embodiment, the lengths **190**, **192** extend outward at an angle between 5 degrees and 45 degrees. The C-hoop **106** can include single unitary bar or rod bent in the shape shown in FIG. **21**. Alternatively, the C-hoop **106** can include multiple segments connected together.

FIG. **22A** is a view of a portion of a sports rack **2200** in raised position including damper **2204** configured as a force transmission **2206** with a force member **2208** for providing powered lifting or human assisted lifting capability, according to an embodiment. FIG. **22B** is a view of the portion of the sports rack **2200** of FIG. **22A** in the lowered position or intermediate position, according to an embodiment.

Referring to FIGS. **22A** and **22B**, a sports rack **2200** for a vehicle **124** may include a hitch bar **112** configured to mount to a vehicle trailer hitch receiver **127**, a hinge assembly **114** operatively coupled to the hitch bar **112**, and at least one post **102** having a proximal end and a distal end. The at least one post **102** may be operatively coupled to the hinge assembly **114** at the proximal end, and the at least one post **102** may be rotatable relative to the hitch bar **112** through an arc in a vertical plane defined by the hinge assembly **114** between a lowered position and a raised position. The sports rack **2200** may include an equipment mount **103** operatively coupled to the distal end of the at least one post **102**, and a damper **152**, **2204** operatively coupled to the at least one post **102** and the hitch bar **112**. The damper **152**, **2204** may be configured to limit a rate of rotation of the at least one post **102** from the raised position to the lowered position.

FIG. **23** is a view of a sports rack **100**, **2200** mounted on a vehicle **124**, the sports rack **100**, **2200** being shown in a raised position, ready for transport, according to an embodiment.

Referring again to FIGS. **22A** and **22B**, in an embodiment, the damper **152**, **2204** is configured to cause the at least one post **102** to rotate from the raised position to the lowered position at a rate slower than a rate of rotation driven by gravity. In an embodiment, the damper **152**, **2204** includes a force transmission **2206** configured to cause the at least one post **102** and the equipment mount **103** to rotate from the lowered position to the raised position. In another embodiment, the damper **152**, **2204** includes a force transmission **2206** configured to urge the at least one post **102** toward the raised position from the lowered position. The force transmission **2206** may be configured to exert sufficient torque to cause the at least one post **102** to rotate from the lowered position to the raised position without human lifting effort when the equipment mount **103** is empty. The force transmission **2206** may be configured to exert sufficient torque to reduce an amount of human lifting effort required or eliminate the human lifting effort altogether to rotate the at least one post **102** from the lowered position to the raised position when the equipment mount **103** is loaded to an equipment lifting capacity. In an embodiment, the force transmission **2206** includes a force multiplier, such as a gear set, a pulley, a cogged pulley, or a block and tackle, for multiplying a force received from a force member **2208**. The force transmission **2206** may include one or more of a lead screw mechanism, a rack and pinion mechanism, or a cogged belt and at least two cogged pulleys. The force transmission **2206** may include a crank for receiving a rotational force from a human user. Alternatively, the force transmission **2206** may include a coupling for receiving a rotational force from a drill or impact driver. In an embodiment, the force transmission **2206** includes a force member **2208** comprising a spring mechanism. In an embodiment, the force transmis-

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sion **2206** includes a force member **2208** comprising a pneumatic cylinder. The force transmission **2206** may include a force member **2208** comprising a winch. Alternatively, the force transmission **2206** may include a force member **2208** comprising a brushless synchronous electric motor. The force transmission **2206** may include a force member **2208** configured to draw electrical current from a trailer wiring socket (not shown).

In an embodiment, the sports rack **2200** further includes a locking mechanism **168** configured to stabilize the at least one post **102** in the raised position. In an embodiment, the sports rack **2200** further includes a detent mechanism or locking mechanism **168** configured to stabilize the at least one post **102** in the lowered position. At least one post **102** may consist essentially of one post **102**. In an embodiment, the at least one post **102** includes a formed post **102** including an off-axis portion disposed to provide a clear field to a vehicle **124** back-up camera (not shown) when the formed post **102** is in the raised position. In another embodiment, the at least one post **102** includes a formed post **102** including an off-axis portion disposed to provide for user access to a vehicle **124** rear hatch control (not shown) when the formed post **102** is in the raised position. Alternatively, the at least one post **102** may include two or more posts **102**, the two or more posts **102** disposed lateral to a centerline of the hitch bar **112**.

In an embodiment, the equipment mount **103** includes a bicycle rack **100** configured to continuously hold one or more bicycles **126a**, **126b** while the at least one post **102** is rotated from the lowered position to the raised position to support the one or more bicycles **126a**, **126b** on the vehicle **124** above ground. The bicycle rack **100** may include a C-hoop **106** coupled to the first support member **104** and having a shape like a letter C.

Referring to FIG. **23**, the equipment mount **103** may include a kayak rack configured to hold one or more respective kayaks **2300**. The equipment mount **103** may include a slot (not shown) configured to receive one or more kayak **2300** T-handles (not shown) disposed near a bow and/or stern (not shown) of respective kayaks **2300**. In an embodiment, the equipment mount **103** includes a mount **106** (such as a C-hoop, as shown) configured to hold either a bicycle wheel **128** or to hook a cockpit of a kayaks **2300**. In an embodiment, the one or more kayaks **2300** include whitewater kayaks **2300**. The one or more kayaks **2300** may be supported in a substantially vertical orientation when the at least one post **102** is in the raised position.

In an embodiment, the bicycle carrier **100** may include integrated locking cables as an anti-theft deterrent when riders are not with the bicycles **126** and bicycle carrier **100**.

Referring to FIGS. **22-23**, the sports rack **100**, **2200** may further include a wall mount bar (see, e.g., wall mount structure **144** of FIG. **9A**) configured to be screwed to a wall (see, e.g., wall **145** of FIG. **9A**), and one or more wall mount fittings (see, e.g., wall mount fittings **149** of FIG. **9B**) configured to be coupled to the wall mount bar and configured to receive and hold a feature of the equipment mount **103**. In an embodiment, the sports rack **100**, **2200** further includes a cable lock loop (see, e.g., cable lock loop **115** of FIG. **1**) coupled to the post **102** or equipment mount **103**. The cable lock loop may be configured to receive one or more locking cables or chains for securing sports equipment to the sports rack **100**, **2200**.

While various aspects and embodiments have been disclosed herein, other aspects and embodiments are contemplated. The various aspects and embodiments disclosed

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herein are for purposes of illustration and are not intended to be limiting, with the true scope and spirit being indicated by the following claims.

What is claimed is:

1. A bicycle carrier with a hitch mount, comprising:
a hitch bar configured to mount to a vehicle trailer hitch receiver;
a hinge assembly operatively coupled to the hitch bar, the hinge assembly including a damper operatively coupled to a support post and the hitch bar, the damper being configured to limit a rate of rotation of the support post relative to the hitch bar;
the support post having a proximal end and a distal end, the support post being operatively coupled to the hinge assembly at the proximal end, the support post being rotatable relative to the hitch bar through an arc in a vertical plane defined by the hinge assembly between a lowered position and a raised position;
a first support member operatively coupled to the distal end of the support post; and
a wheel-receiving member coupled to the first support member, the wheel-receiving member being formed and configured to receive a wheel of a bicycle such that the bicycle is suspended vertically while the wheel is in the wheel-receiving member, the wheel-receiving member being further formed and configured to receive the wheel at any wheel angle within a continuous range of wheel angles with respect to the support post, the range of wheel angles including at least all angles between and including an angle substantially parallel to the support post and an acute angle with respect to the support post.
2. The bicycle carrier with the hitch mount of claim 1, further comprising a locking latch pin that, when engaged, is configured to prevent rotation of the support post.
3. The bicycle carrier with the hitch mount of claim 2, wherein the locking latch pin is configured to hold the support post in a plurality of selected positions including at least the lowered loading stop position and the raised transport position.
4. The bicycle carrier with the hitch mount of claim 3, wherein in the raised transport position, the support post is oriented vertically.
5. The bicycle carrier with the hitch mount of claim 4, wherein, when the support post is in the vertical position, the wheel receiving member defines a plane substantially parallel to the ground and perpendicular to the vertical direction.
6. The bicycle carrier with the hitch mount of claim 3, wherein in the lowered loading stop position, the support post is oriented at about 33° from vertical.
7. The bicycle carrier with the hitch mount of claim 2, further comprising a hinge release lever operatively coupled to the locking latch pin and configured to release the locking latch pin when pulled by a user.
8. The bicycle carrier with the hitch mount of claim 1, further comprising a hinge stop pin configured to define angles to which the support post is rotatable.
9. The bicycle carrier with the hitch mount of claim 1, wherein the damper includes a fluid chamber and a piston, wherein fluid flow into or out of the fluid chamber is inhibited by an aperture.
10. The bicycle carrier with the hitch mount of claim 1, wherein the damper includes a spring.
11. The bicycle carrier with the hitch mount of claim 10, wherein the spring includes a gas spring.

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12. The bicycle carrier with the hitch mount of claim 1, wherein the damper is pneumatic or hydraulic.

13. The bicycle carrier with the hitch mount of claim 1, wherein the wheel receiving member is configured to support the wheel by contact with a tire of the wheel without contacting spokes of the wheel or frame of the bicycle.

14. The bicycle carrier with the hitch mount of claim 13, wherein, after being positioned in a wheel receiving member, the wheel of the bicycle comes to rest at an angle other than vertical.

15. The bicycle carrier with the hitch mount of claim 14, wherein the wheel of the bicycle rests against an angled support member when the wheel of the bicycle comes to rest.

16. The bicycle carrier with the hitch mount of claim 1, wherein the wheel receiving member is configured to receive the wheel at a wheel angle substantially parallel to the support post and to then permit the wheel to rotate to a wheel angle of at least 20 degrees relative to the support post.

17. The bicycle carrier with the hitch mount of claim 1, wherein the wheel receiving member is about 26 inches long, perpendicular to the first support member.

18. The bicycle carrier with the hitch mount of claim 1, wherein the bicycle is suspended vertically while the wheel is in the wheel-receiving member when the support post is in the raised position.

19. The bicycle carrier with the hitch mount of claim 1, wherein the wheel receiving member is in the form of a "C".

20. The bicycle carrier with the hitch mount of claim 1, wherein the wheel-receiving member is coupled to the first support member and supported only by a coupling member disposed between the wheel-receiving member and the first support member, the coupling member extending from only one side of the wheel-receiving member.

21. The bicycle carrier with the hitch mount of claim 20, wherein the coupling member is angled relative to the first support member at an angle greater than 20 degrees from vertical when the support post is in the raised position.

22. The bicycle carrier with the hitch mount of claim 1, wherein the wheel receiving member includes a plurality of wheel receiving members configured to receive wheels of a corresponding plurality of bicycles.

23. The bicycle carrier with the hitch mount of claim 22, wherein the plurality of wheel receiving members includes seven wheel receiving members.

24. The bicycle carrier with the hitch mount of claim 22, wherein the plurality of wheel receiving members are spaced across the first support member to support the plurality of bicycles without mechanical interference between the plurality of bicycles.

25. A bicycle carrier with a hitch mount, comprising:
a hitch bar configured to mount to a vehicle trailer hitch receiver;
a hinge assembly operatively coupled to the hitch bar, the hinge assembly including a damper operatively coupled to a support post and the hitch bar, the damper being configured to limit a rate of rotation of the support post relative to the hitch bar;
the support post having a proximal end and a distal end, the support post being operatively coupled to the hinge assembly at the proximal end, the support post being rotatable relative to the hitch bar through an arc in a vertical plane defined by the hinge assembly between a lowered position and a raised position;
a first support member operatively coupled to the distal end of the support post; and

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a wheel-receiving member coupled to the first support member, the wheel-receiving member being configured to receive a wheel of a bicycle such that the bicycle is suspended vertically while the wheel is in the wheel-receiving member, the wheel-receiving member being 5 further configured to receive the wheel at any of a range of angles with respect to the support post, the range of angles including at least an angle substantially parallel to the support post and an acute angle with respect to the support post, 10 wherein the damper is disposed in a retracted state while the support post is in the raised position.

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